

MIT Academy of Engineering , Alandi

DATA VISUALIZATION MINI PROJECT (DV)

MOVIE RECOMMENDATION SYSTEM

Course Code: 2311162L

Course Name: Data Visualization Lab

Academic Year: 2025-26

Student Details :

- **Student Name:** Sukrut Ujwal Dusane
- **PRN Number:** 202501110114
- **Division / Batch:** AL2 - A21
- **Dataset Name & Source:** Netflix Tv shows and Movies
<https://www.kaggle.com/datasets/victorsoeiro/netflix-tv-shows-and-movies>

Title : Netflix Content-Based Movie Recommendation System Using Machine Learning

Objective: The objective of this project is to design and develop an intelligent movie recommendation system using the Netflix Movies and TV Shows dataset. The system aims to analyze content features such as genre, description, cast, directors, IMDb ratings, runtime, and release year to identify similarities between titles and recommend relevant movies or TV shows to users based on their preferences. By applying data preprocessing, feature engineering, and similarity-based machine learning techniques, the project seeks to improve user content discovery and enhance personalized entertainment experiences. The dataset contains over 5,000 Netflix titles and 77,000+ cast/credit records, making it suitable for building a rich recommendation engine.

Problem Statement : With the rapid growth of streaming platforms like Netflix, users are exposed to thousands of content options, making it difficult to find movies or shows that match their interests. Traditional browsing methods often lead to decision fatigue and reduced user satisfaction. The challenge is to build an efficient recommendation system that can automatically suggest relevant content based on metadata such as genres, descriptions, cast information, and popularity scores. Using the Netflix dataset collected from JustWatch, this project addresses the problem by implementing a content-based filtering model that analyzes similarities among titles and provides personalized recommendations, helping users discover content more effectively while improving engagement on streaming platforms.

Project Requirements :

1. Data Loading :

The first step of the project involves loading the Netflix Movies and TV Shows dataset into Power BI from Kaggle. The dataset is imported in CSV format and verified for structure, data types, and completeness to ensure it is ready for further analysis.

credits.csv

File Origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

person_id	id	name	character	role
3748	tm84618	Robert De Niro	Travis Bickle	ACTOR
14658	tm84618	Jodie Foster	Iris Steensma	ACTOR
7064	tm84618	Albert Brooks	Tom	ACTOR
3739	tm84618	Harvey Keitel	Matthew 'Sport' Higgins	ACTOR
48933	tm84618	Cybill Shepherd	Betsy	ACTOR
32267	tm84618	Peter Boyle	Wizard	ACTOR
519612	tm84618	Leonard Harris	Senator Charles Palantine	ACTOR
29068	tm84618	Diahnne Abbott	Concession Girl	ACTOR
519613	tm84618	Gino Ardito	Policeman at Rally	ACTOR
3308	tm84618	Martin Scorsese	Passenger Watching Silhouette	ACTOR
43791	tm84618	Murray Moston	Iris' Time Keeper	ACTOR
519614	tm84618	Richard Higgs	Secret Service Agent	ACTOR
519615	tm84618	Bill Minkin	Tom's Assistant (uncredited)	ACTOR
82426	tm84618	Bob Maroff	Mafioso (uncredited)	ACTOR
20935	tm84618	Victor Argo	Melio, Delicatessen Owner	ACTOR
7753	tm84618	Joe Spinell	Personell Officer	ACTOR
43279	tm84618	Robinson Frank Adu	Angry Black Man (uncredited)	ACTOR
519616	tm84618	Brenda Dickson	Soap Opera Woman (uncredited)	ACTOR
8424	tm84618	Norman Matlock	Charlie T	ACTOR
20447	tm84618	Harry Northup	Doughboy	ACTOR

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

titles.csv

File Origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

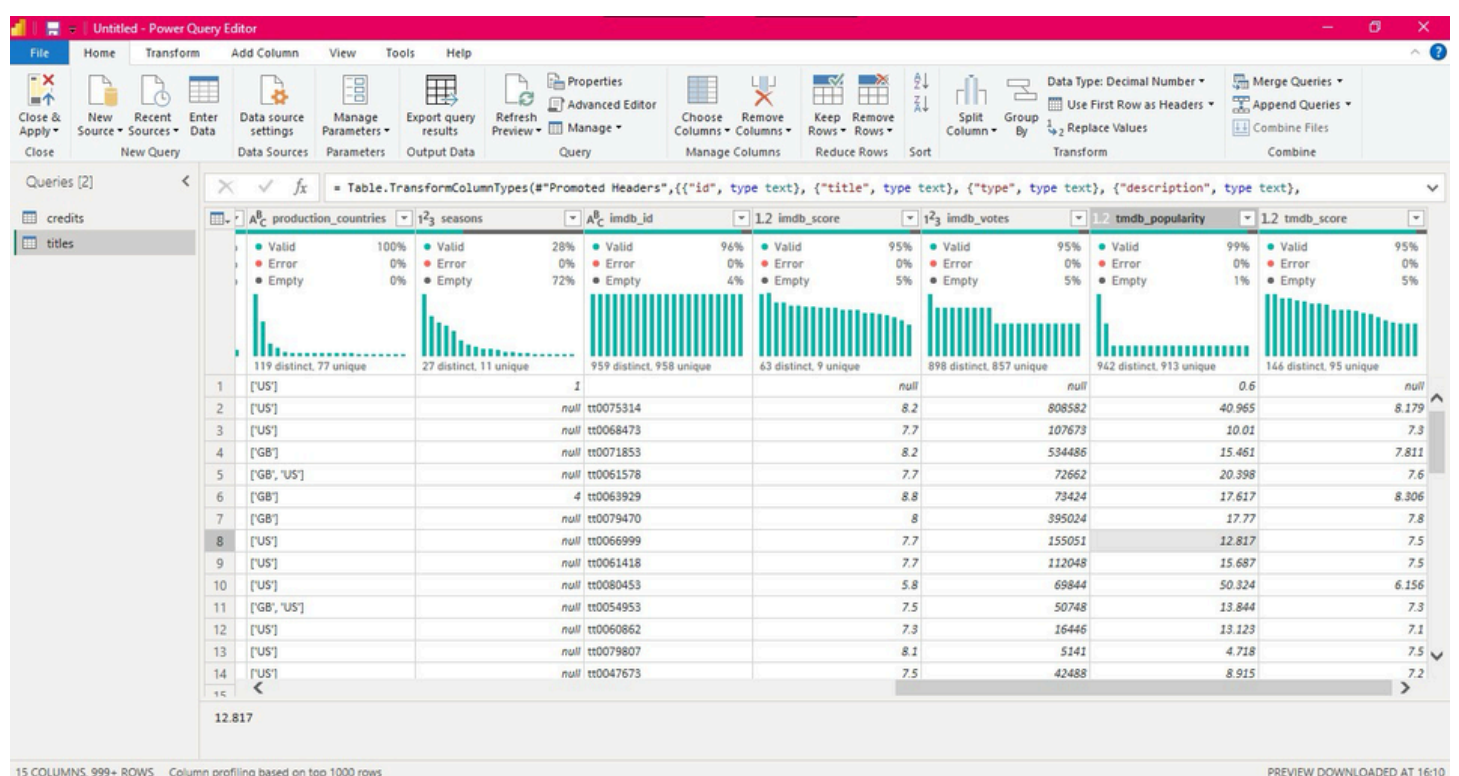
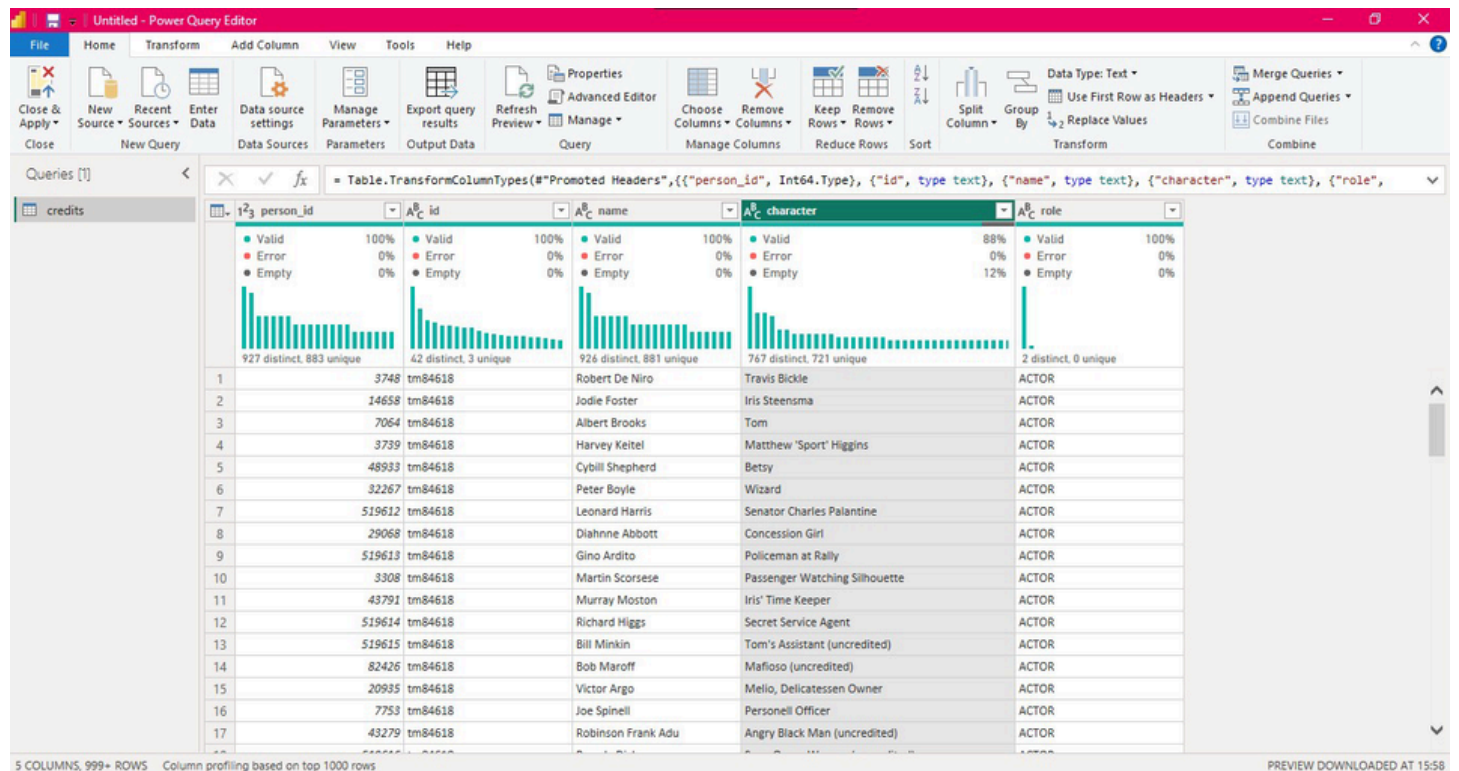
id	title	type	description	release_year	age_certification	run
ts300399	Five Came Back: The Reference Films	SHOW	This collection includes 12 World War II-era propagand...	1945	TV-MA	
tm84618	Taxi Driver	MOVIE	A mentally unstable Vietnam War veteran works as a ni...	1976	R	
tm154986	Deliverance	MOVIE	Intent on seeing the Cahulawassee River before it's tur...	1972	R	
tm127384	Monty Python and the Holy Grail	MOVIE	King Arthur, accompanied by his squire, recruits his kni...	1975	PG	
tm120801	The Dirty Dozen	MOVIE	12 American military prisoners in World War II are orde...	1967		
ts22164	Monty Python's Flying Circus	SHOW	A British sketch comedy series with the shows being co...	1969	TV-14	
tm70993	Life of Brian	MOVIE	Brian Cohen is an average young Jewish man, but throu...	1979	R	
tm14873	Dirty Harry	MOVIE	When a madman dubbed 'Scorpio' terrorizes San Franci...	1971	R	
tm119281	Bonnie and Clyde	MOVIE	In the 1930s, bored waitress Bonnie Parker falls in love...	1967	R	
tm98978	The Blue Lagoon	MOVIE	Two small children and a ship's cook survive a shipwrec...	1980	R	
tm44204	The Guns of Navarone	MOVIE	A team of allied saboteurs are assigned an impossible...	1961		
tm67378	The Professionals	MOVIE	An arrogant Texas millionaire hires four adventurers to...	1966	PG-13	
tm69997	Richard Pryor: Live in Concert	MOVIE	Richard Pryor delivers monologues on race, sex, family...	1979	R	
tm16479	White Christmas	MOVIE	Two talented song-and-dance men team up after the w...	1954		
tm135083	Cairo Station	MOVIE	Qinawi, a physically challenged peddler who makes his l...	1958		
tm89386	Hitler: A Career	MOVIE	A keen chronicle of the unlikely rise to power of Adolf...	1977	PG	
tm156453	FTA	MOVIE	A documentary about a political troupe headed by acto...	1972	R	
tm14350	Alexandria... Why?	MOVIE	Set against the panoramic backdrop of war-torn Egypt...	1979		
tm81728	The Land	MOVIE	Set in 1933, the mayor informs the peasants that the sh...	1969		
tm94651	Dostana	MOVIE	Two close friends decide to enter law enforcement, on...	1980		

Extract Table Using Examples | Load | Transform Data | Cancel

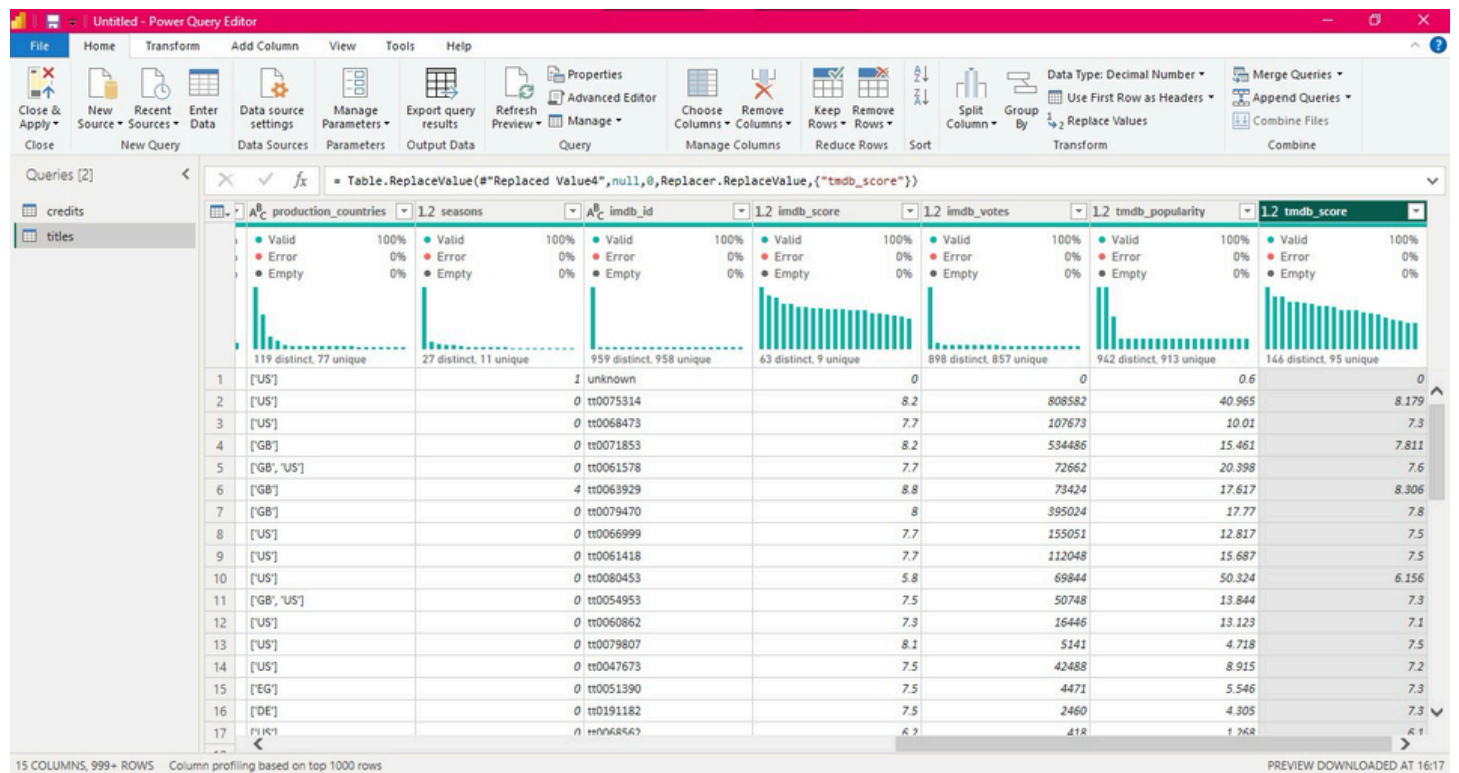
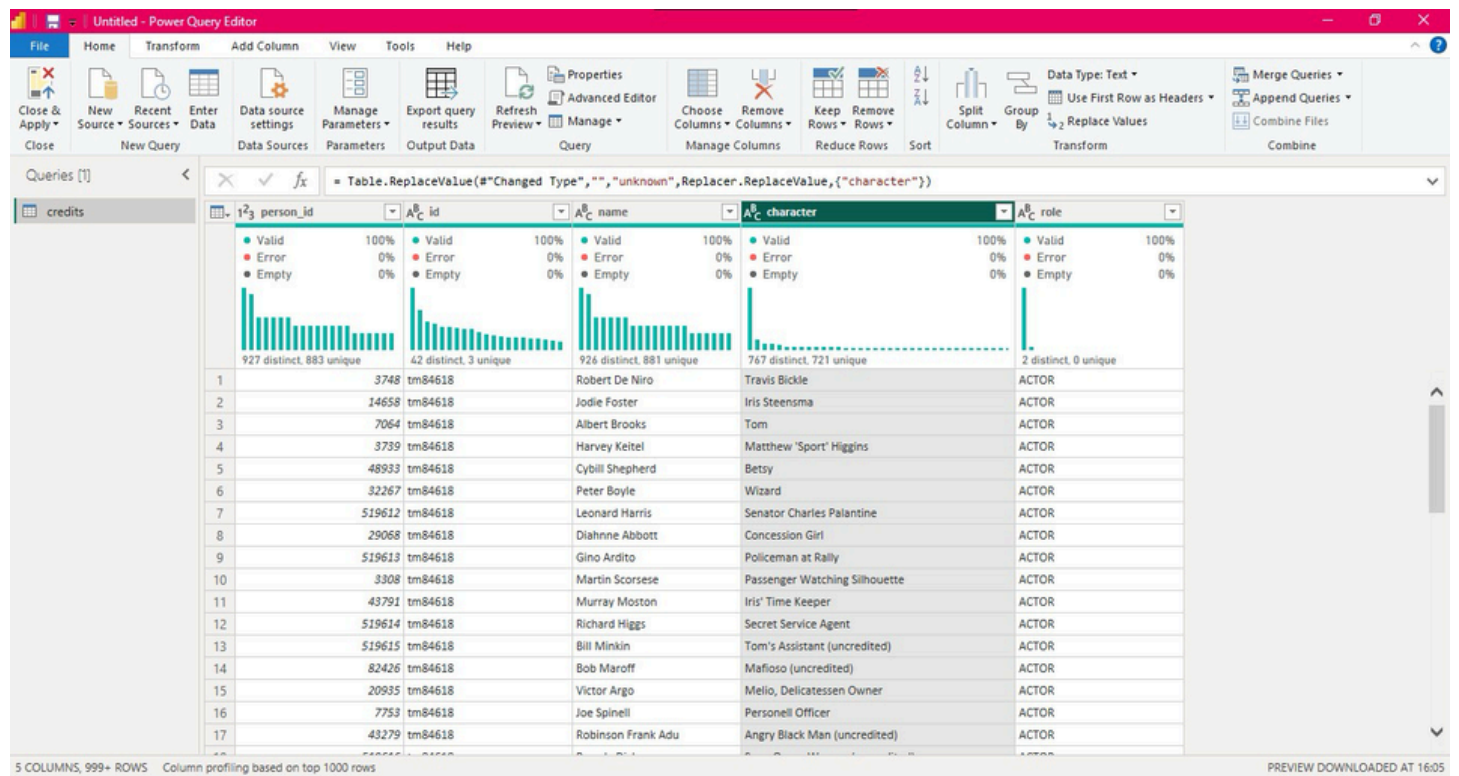
2. Data Transformation (Cleaning & Preprocessing)

Data transformation includes removing duplicate records, handling null values, correcting inconsistent formats, splitting multi-valued columns like genres and cast, and converting date fields into proper date formats. This improves data quality and reliability for analysis.

Before Data Cleaning -



After Data Cleaning -



Data Cleaning (By replacing null values) -

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [1] credits

Table.ReplaceValue(#"Changed Type",null,"unknown",Replacer.ReplaceValue,{"character"})

person_id id name character role

Valid 100% Error 0% Empty 0%

7 distinct, 883 unique

Replace Values

Replace one value with another in the selected columns.

Value To Find

Replace With

unknown

Advanced options

OK Cancel

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 16:01

Query Settings

PROPERTIES

Name credits

APPLIED STEPS

Source Promoted Headers Changed Type Replaced Value

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [2] credits titles

Table.TransformColumnTypes(#"Promoted Headers",{{"id", type text}, {"title", type text}, {"type", type text}, {"description", type text}, {"production_countries", type text}, {"seasons", type text}, {"imdb_id", type text}, {"imdb_score", type text}, {"imdb_votes", type text}, {"tmdb_popularity", type text}, {"tmdb_score", type text}}

production_countries seasons imdb_id imdb_score imdb_votes tmdb_popularity tmdb_score

Valid 100% Error 0% Empty 0%

119 distinct, 77 unique

Replace Values

Replace one value with another in the selected columns.

Value To Find

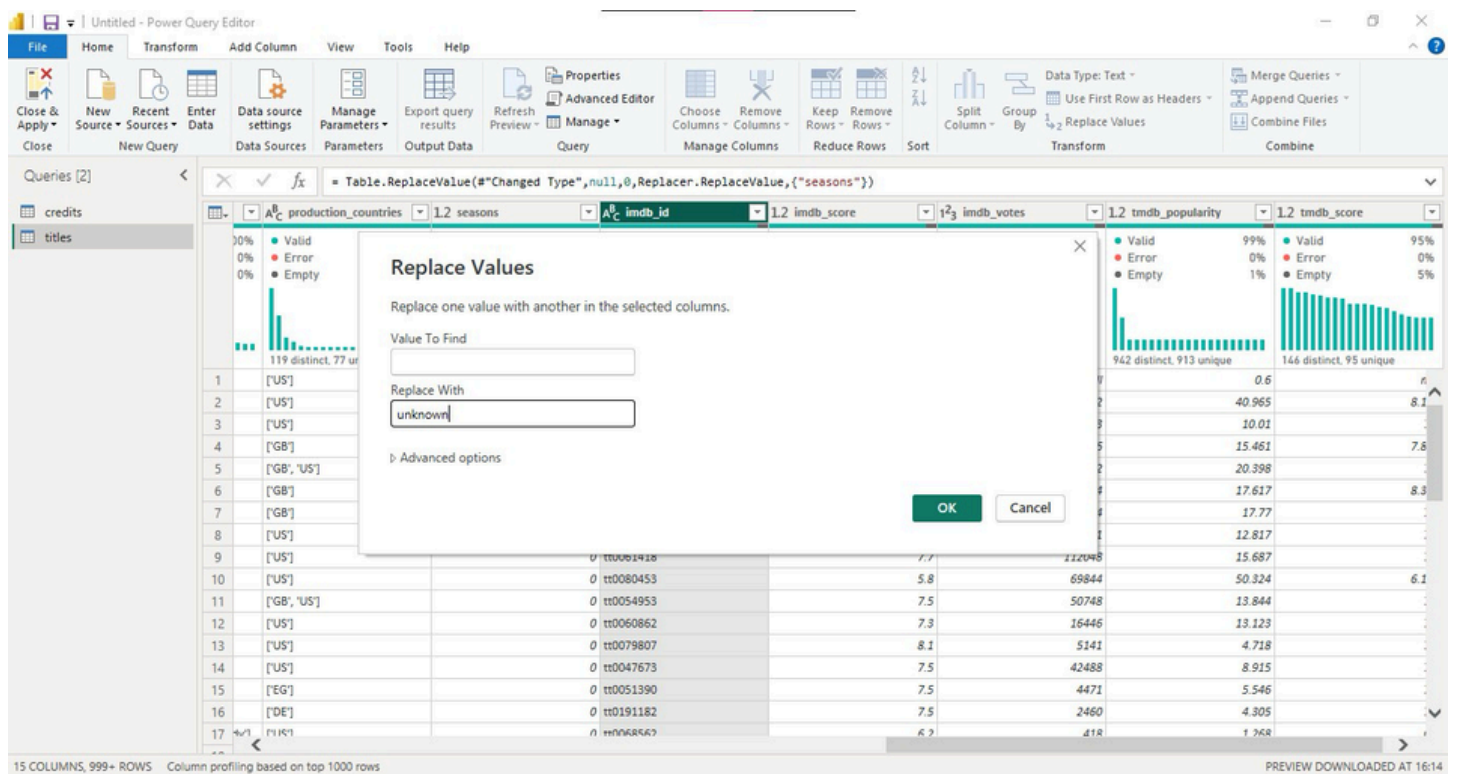
Replace With

d

OK Cancel

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 16:10



3. Creating Data Model (relationships)

In this project, a structured data model is created by establishing relationships between the main **Fact_Titles** table and supporting dimension tables such as **Dim_Credits** and **Calendar** in Power BI. The **Fact_Titles** table contains the primary Netflix content information, while the **Dim_Credits** table stores cast and crew details linked through a common content ID, and the **Calendar** table enables time-based analysis using release dates. These relationships follow a **star schema model**, which improves data organization, enables efficient filtering across tables, and supports accurate analysis for visualizations such as genre trends, yearly content additions, top actors, and recommendation insights. This relational model ensures better performance, consistency, and scalability in the dashboard.

New relationship
Select tables and columns that are related.

From table: credits

character	id	name	person_id	role
unknown	tm84618	Martin Scorse...	3308	DIRECTOR
unknown	tm154986	John Boorman	17727	DIRECTOR
unknown	tm127384	Terry Jones	11475	DIRECTOR

To table: titles

age_certificati...	description	genres	id	imdb_id	imdb_score	imdb_vote
R	Sane Man wa...	[comedy]	tm132164	unknown	0	0
	Get ready to s...	[comedy]	tm8687	unknown	0	0
NC-17	Bill Hicks sho...	[comedy]	tm7135	unknown	0	0

Cardinality: Many to one (*:1)
Cross-filter direction: Single

☒ Make this relationship active
☐ Assume referential integrity
☐ Apply security filter in both directions

There's already a relationship between these two columns.

Save Cancel

New relationship
Select tables and columns that are related.

From table: Fact_titles

release_year	runtime	seasons	title	tmdb_popula...	tmdb_score	type
1989	80	0	Bill Hicks: San...	3.377	7.5	MOVIE
1991	49	0	Sam Kinison: ...	1.84	6.1	MOVIE
1991	30	0	Bill Hicks: On...	1.08	7.5	MOVIE

To table: Calendar

Date	Month	Year
01-01-1950 0...	January	1950
02-01-1950 0...	January	1950
03-01-1950 0...	January	1950

Cardinality: Many to many (*:*)
Cross-filter direction: Both

☒ Make this relationship active
☐ Assume referential integrity
☐ Apply security filter in both directions

There's already a relationship between these two columns.

Save Cancel

4. Creation of Custom Calendar Table

In this project, a custom **Calendar Table** is created in Power BI to support efficient time-based analysis of Netflix content data. The table is generated using DAX by creating a continuous date range based on the release dates available in the dataset. Additional calculated columns such as **Year**, **Month**, **Quarter**, and **Day** are added to simplify filtering and trend analysis. This calendar table is connected to the main **Fact_Titles** table through the

release year/date field, enabling the use of time intelligence functions for analyzing yearly content growth, monthly release patterns, and content trends over time. The custom calendar table improves data organization and makes temporal analysis more accurate and efficient.

Untitled - Power BI Desktop

File Home Help Table tools

Name Calendar

Manage relationships Relationships

New measure Quick measure New column New table Calculations

Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Calendar = CALENDAR(DATE(1950,1,1),DATE(2026,12,31))

Date

01-01-1950 00:00:00
02-01-1950 00:00:00
03-01-1950 00:00:00
04-01-1950 00:00:00
05-01-1950 00:00:00
06-01-1950 00:00:00
07-01-1950 00:00:00
08-01-1950 00:00:00
09-01-1950 00:00:00
10-01-1950 00:00:00
11-01-1950 00:00:00
12-01-1950 00:00:00
13-01-1950 00:00:00
14-01-1950 00:00:00
15-01-1950 00:00:00
16-01-1950 00:00:00
17-01-1950 00:00:00
18-01-1950 00:00:00
19-01-1950 00:00:00
20-01-1950 00:00:00
21-01-1950 00:00:00
22-01-1950 00:00:00
23-01-1950 00:00:00
24-01-1950 00:00:00
25-01-1950 00:00:00
26-01-1950 00:00:00
27-01-1950 00:00:00
28-01-1950 00:00:00
29-01-1950 00:00:00
30-01-1950 00:00:00
31-01-1950 00:00:00
01-02-1950 00:00:00
02-02-1950 00:00:00
03-02-1950 00:00:00
04-02-1950 00:00:00
05-02-1950 00:00:00
06-02-1950 00:00:00
07-02-1950 00:00:00
08-02-1950 00:00:00
09-02-1950 00:00:00
10-02-1950 00:00:00
11-02-1950 00:00:00
12-02-1950 00:00:00
13-02-1950 00:00:00
14-02-1950 00:00:00
15-02-1950 00:00:00
16-02-1950 00:00:00
17-02-1950 00:00:00
18-02-1950 00:00:00
19-02-1950 00:00:00
20-02-1950 00:00:00
21-02-1950 00:00:00
22-02-1950 00:00:00
23-02-1950 00:00:00
24-02-1950 00:00:00
25-02-1950 00:00:00
26-02-1950 00:00:00
27-02-1950 00:00:00
28-02-1950 00:00:00
29-02-1950 00:00:00
01-03-1950 00:00:00
02-03-1950 00:00:00
03-03-1950 00:00:00
04-03-1950 00:00:00
05-03-1950 00:00:00
06-03-1950 00:00:00
07-03-1950 00:00:00
08-03-1950 00:00:00
09-03-1950 00:00:00
10-03-1950 00:00:00
11-03-1950 00:00:00
12-03-1950 00:00:00
13-03-1950 00:00:00
14-03-1950 00:00:00
15-03-1950 00:00:00
16-03-1950 00:00:00
17-03-1950 00:00:00
18-03-1950 00:00:00
19-03-1950 00:00:00
20-03-1950 00:00:00
21-03-1950 00:00:00
22-03-1950 00:00:00
23-03-1950 00:00:00
24-03-1950 00:00:00
25-03-1950 00:00:00
26-03-1950 00:00:00
27-03-1950 00:00:00
28-03-1950 00:00:00
29-03-1950 00:00:00
30-03-1950 00:00:00
31-03-1950 00:00:00
01-04-1950 00:00:00
02-04-1950 00:00:00
03-04-1950 00:00:00
04-04-1950 00:00:00
05-04-1950 00:00:00
06-04-1950 00:00:00
07-04-1950 00:00:00
08-04-1950 00:00:00
09-04-1950 00:00:00
10-04-1950 00:00:00
11-04-1950 00:00:00
12-04-1950 00:00:00
13-04-1950 00:00:00
14-04-1950 00:00:00
15-04-1950 00:00:00
16-04-1950 00:00:00
17-04-1950 00:00:00
18-04-1950 00:00:00
19-04-1950 00:00:00
20-04-1950 00:00:00
21-04-1950 00:00:00
22-04-1950 00:00:00
23-04-1950 00:00:00
24-04-1950 00:00:00
25-04-1950 00:00:00
26-04-1950 00:00:00
27-04-1950 00:00:00
28-04-1950 00:00:00
29-04-1950 00:00:00
30-04-1950 00:00:00
01-05-1950 00:00:00
02-05-1950 00:00:00
03-05-1950 00:00:00
04-05-1950 00:00:00
05-05-1950 00:00:00
06-05-1950 00:00:00
07-05-1950 00:00:00
08-05-1950 00:00:00
09-05-1950 00:00:00
10-05-1950 00:00:00
11-05-1950 00:00:00
12-05-1950 00:00:00
13-05-1950 00:00:00
14-05-1950 00:00:00
15-05-1950 00:00:00
16-05-1950 00:00:00
17-05-1950 00:00:00
18-05-1950 00:00:00
19-05-1950 00:00:00
20-05-1950 00:00:00
21-05-1950 00:00:00
22-05-1950 00:00:00
23-05-1950 00:00:00
24-05-1950 00:00:00
25-05-1950 00:00:00
26-05-1950 00:00:00
27-05-1950 00:00:00
28-05-1950 00:00:00
29-05-1950 00:00:00
30-05-1950 00:00:00
31-05-1950 00:00:00
01-06-1950 00:00:00
02-06-1950 00:00:00
03-06-1950 00:00:00
04-06-1950 00:00:00
05-06-1950 00:00:00
06-06-1950 00:00:00
07-06-1950 00:00:00
08-06-1950 00:00:00
09-06-1950 00:00:00
10-06-1950 00:00:00
11-06-1950 00:00:00
12-06-1950 00:00:00
13-06-1950 00:00:00
14-06-1950 00:00:00
15-06-1950 00:00:00
16-06-1950 00:00:00
17-06-1950 00:00:00
18-06-1950 00:00:00
19-06-1950 00:00:00
20-06-1950 00:00:00
21-06-1950 00:00:00
22-06-1950 00:00:00
23-06-1950 00:00:00
24-06-1950 00:00:00
25-06-1950 00:00:00
26-06-1950 00:00:00
27-06-1950 00:00:00
28-06-1950 00:00:00
29-06-1950 00:00:00
30-06-1950 00:00:00
01-07-1950 00:00:00
02-07-1950 00:00:00
03-07-1950 00:00:00
04-07-1950 00:00:00
05-07-1950 00:00:00
06-07-1950 00:00:00
07-07-1950 00:00:00
08-07-1950 00:00:00
09-07-1950 00:00:00
10-07-1950 00:00:00
11-07-1950 00:00:00
12-07-1950 00:00:00
13-07-1950 00:00:00
14-07-1950 00:00:00
15-07-1950 00:00:00
16-07-1950 00:00:00
17-07-1950 00:00:00
18-07-1950 00:00:00
19-07-1950 00:00:00
20-07-1950 00:00:00
21-07-1950 00:00:00
22-07-1950 00:00:00
23-07-1950 00:00:00
24-07-1950 00:00:00
25-07-1950 00:00:00
26-07-1950 00:00:00
27-07-1950 00:00:00
28-07-1950 00:00:00
29-07-1950 00:00:00
30-07-1950 00:00:00
31-07-1950 00:00:00
01-08-1950 00:00:00
02-08-1950 00:00:00
03-08-1950 00:00:00
04-08-1950 00:00:00
05-08-1950 00:00:00
06-08-1950 00:00:00
07-08-1950 00:00:00
08-08-1950 00:00:00
09-08-1950 00:00:00
10-08-1950 00:00:00
11-08-1950 00:00:00
12-08-1950 00:00:00
13-08-1950 00:00:00
14-08-1950 00:00:00
15-08-1950 00:00:00
16-08-1950 00:00:00
17-08-1950 00:00:00
18-08-1950 00:00:00
19-08-1950 00:00:00
20-08-1950 00:00:00
21-08-1950 00:00:00
22-08-1950 00:00:00
23-08-1950 00:00:00
24-08-1950 00:00:00
25-08-1950 00:00:00
26-08-1950 00:00:00
27-08-1950 00:00:00
28-08-1950 00:00:00
29-08-1950 00:00:00
30-08-1950 00:00:00
31-08-1950 00:00:00
01-09-1950 00:00:00
02-09-1950 00:00:00
03-09-1950 00:00:00
04-09-1950 00:00:00
05-09-1950 00:00:00
06-09-1950 00:00:00
07-09-1950 00:00:00
08-09-1950 00:00:00
09-09-1950 00:00:00
10-09-1950 00:00:00
11-09-1950 00:00:00
12-09-1950 00:00:00
13-09-1950 00:00:00
14-09-1950 00:00:00
15-09-1950 00:00:00
16-09-1950 00:00:00
17-09-1950 00:00:00
18-09-1950 00:00:00
19-09-1950 00:00:00
20-09-1950 00:00:00
21-09-1950 00:00:00
22-09-1950 00:00:00
23-09-1950 00:00:00
24-09-1950 00:00:00
25-09-1950 00:00:00
26-09-1950 00:00:00
27-09-1950 00:00:00
28-09-1950 00:00:00
29-09-1950 00:00:00
30-09-1950 00:00:00
01-10-1950 00:00:00
02-10-1950 00:00:00
03-10-1950 00:00:00
04-10-1950 00:00:00
05-10-1950 00:00:00
06-10-1950 00:00:00
07-10-1950 00:00:00
08-10-1950 00:00:00
09-10-1950 00:00:00
10-10-1950 00:00:00
11-10-1950 00:00:00
12-10-1950 00:00:00
13-10-1950 00:00:00
14-10-1950 00:00:00
15-10-1950 00:00:00
16-10-1950 00:00:00
17-10-1950 00:00:00
18-10-1950 00:00:00
19-10-1950 00:00:00
20-10-1950 00:00:00
21-10-1950 00:00:00
22-10-1950 00:00:00
23-10-1950 00:00:00
24-10-1950 00:00:00
25-10-1950 00:00:00
26-10-1950 00:00:00
27-10-1950 00:00:00
28-10-1950 00:00:00
29-10-1950 00:00:00
30-10-1950 00:00:00
31-10-1950 00:00:00
01-11-1950 00:00:00
02-11-1950 00:00:00
03-11-1950 00:00:00
04-11-1950 00:00:00
05-11-1950 00:00:00
06-11-1950 00:00:00
07-11-1950 00:00:00
08-11-1950 00:00:00
09-11-1950 00:00:00
10-11-1950 00:00:00
11-11-1950 00:00:00
12-11-1950 00:00:00
13-11-1950 00:00:00
14-11-1950 00:00:00
15-11-1950 00:00:00
16-11-1950 00:00:00
17-11-1950 00:00:00
18-11-1950 00:00:00
19-11-1950 00:00:00
20-11-1950 00:00:00
21-11-1950 00:00:00
22-11-1950 00:00:00
23-11-1950 00:00:00
24-11-1950 00:00:00
25-11-1950 00:00:00
26-11-1950 00:00:00
27-11-1950 00:00:00
28-11-1950 00:00:00
29-11-1950 00:00:00
30-11-1950 00:00:00
01-12-1950 00:00:00
02-12-1950 00:00:00
03-12-1950 00:00:00
04-12-1950 00:00:00
05-12-1950 00:00:00
06-12-1950 00:00:00
07-12-1950 00:00:00
08-12-1950 00:00:00
09-12-1950 00:00:00
10-12-1950 00:00:00
11-12-1950 00:00:00
12-12-1950 00:00:00
13-12-1950 00:00:00
14-12-1950 00:00:00
15-12-1950 00:00:00
16-12-1950 00:00:00
17-12-1950 00:00:00
18-12-1950 00:00:00
19-12-1950 00:00:00
20-12-1950 00:00:00
21-12-1950 00:00:00
22-12-1950 00:00:00
23-12-1950 00:00:00
24-12-1950 00:00:00
25-12-1950 00:00:00
26-12-1950 00:00:00
27-12-1950 00:00:00
28-12-1950 00:00:00
29-12-1950 00:00:00
30-12-1950 00:00:00
31-12-1950 00:00:00

Table: Calendar (28,124 rows)

Update available (click to download)

Search

Calendar

Dim_credits

- character
- id
- name
- person_id
- role

Fact_titles

- age_certification
- description
- genres
- id
- imdb_id
- imdb_score
- imdb_votes
- production_countries
- release_year
- runtime
- seasons

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Year

Data type Whole number

Format Whole number

Summarization Sum

Data category Uncategorized

Sort by column

Data groups

Manage relationships

New column

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Year = YEAR('Calendar'[Date])

Date Year

01-01-1950 00:00:00	1950
02-01-1950 00:00:00	1950
03-01-1950 00:00:00	1950
04-01-1950 00:00:00	1950
05-01-1950 00:00:00	1950
06-01-1950 00:00:00	1950
07-01-1950 00:00:00	1950
08-01-1950 00:00:00	1950
09-01-1950 00:00:00	1950
10-01-1950 00:00:00	1950
11-01-1950 00:00:00	1950
12-01-1950 00:00:00	1950
13-01-1950 00:00:00	1950
14-01-1950 00:00:00	1950
15-01-1950 00:00:00	1950
16-01-1950 00:00:00	1950
17-01-1950 00:00:00	1950
18-01-1950 00:00:00	1950
19-01-1950 00:00:00	1950
20-01-1950 00:00:00	1950
21-01-1950 00:00:00	1950
22-01-1950 00:00:00	1950
23-01-1950 00:00:00	1950
24-01-1950 00:00:00	1950
25-01-1950 00:00:00	1950
26-01-1950 00:00:00	1950
27-01-1950 00:00:00	1950
28-01-1950 00:00:00	1950
29-01-1950 00:00:00	1950
30-01-1950 00:00:00	1950
31-01-1950 00:00:00	1950
01-02-1950 00:00:00	1950
02-02-1950 00:00:00	1950
03-02-1950 00:00:00	1950
04-02-1950 00:00:00	1950
05-02-1950 00:00:00	1950
06-02-1950 00:00:00	1950
07-02-1950 00:00:00	1950
08-02-1950 00:00:00	1950
09-02-1950 00:00:00	1950
10-02-1950 00:00:00	1950
11-02-1950 00:00:00	1950
12-02-1950 00:00:00	

Untitled - Power BI Desktop

Search

Sign in

File Home Help Table tools Column tools

Name: Month

Data type: Text

Format: Text

Summarization: Don't summarize

Data category: Uncategorized

Sort by column: Sort

Data groups: Groups

Manage relationships: Relationships

New column: Calculations

Structure

Formatting

Properties

Auto recovery contains some recovered files that haven't been opened.

View recovered files

Data

1 Month = FORMAT('Calendar'[Date], "MMMM")

Date	Year	Month
01-01-1950 00:00:00	1950	January
02-01-1950 00:00:00	1950	January
03-01-1950 00:00:00	1950	January
04-01-1950 00:00:00	1950	January
05-01-1950 00:00:00	1950	January
06-01-1950 00:00:00	1950	January
07-01-1950 00:00:00	1950	January
08-01-1950 00:00:00	1950	January
09-01-1950 00:00:00	1950	January
10-01-1950 00:00:00	1950	January
11-01-1950 00:00:00	1950	January
12-01-1950 00:00:00	1950	January
13-01-1950 00:00:00	1950	January
14-01-1950 00:00:00	1950	January
15-01-1950 00:00:00	1950	January
16-01-1950 00:00:00	1950	January
17-01-1950 00:00:00	1950	January
18-01-1950 00:00:00	1950	January
19-01-1950 00:00:00	1950	January
20-01-1950 00:00:00	1950	January
21-01-1950 00:00:00	1950	January
22-01-1950 00:00:00	1950	January

Table: Calendar (28,124 rows) Column: Month (12 distinct values)

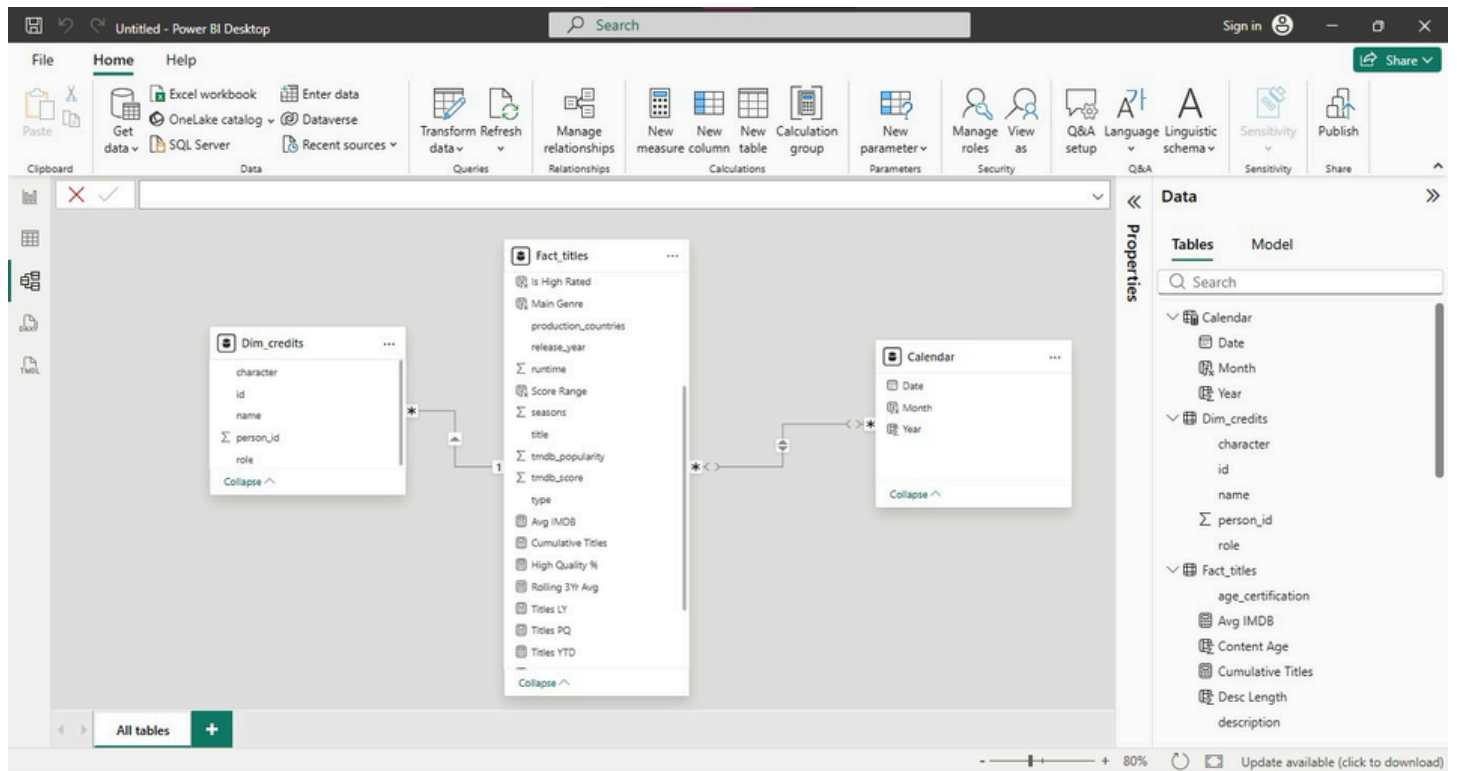
Update available (click to download)

Search

- Calendar
 - Date
 - Month
 - Year
- Dim_credits
 - character
 - id
 - name
 - person_id
 - role
- Fact_titles
 - age_certification
 - description
 - genres
 - id
 - imdb_id
 - imdb_score
 - imdb_votes
 - production_countries

5. Preparation of Schema Diagram (Star / Snowflake)

In this project, a **Star Schema** is designed to organize the Netflix dataset efficiently for analysis in Power BI. The central **Fact_Titles** table acts as the fact table containing the core content details such as title, genre, runtime, IMDb score, popularity, and release year. It is connected to dimension tables like **Dim_Credits** (cast and crew details) and **Calendar** (date-related information) through one-to-many relationships. This schema structure simplifies data management, improves query performance, and enables faster filtering and aggregation across related tables. By using the star schema model, the dashboard becomes more scalable, easier to understand, and better suited for generating meaningful insights and recommendation analysis.



6. Writing Grain (level of data detail)

1. What is the total number of titles available on Netflix?
2. What is the average IMDb rating of Netflix content?
3. How many movies and TV shows are available on Netflix?
4. Which genre has the highest number of titles?
5. How has Netflix content grown over the years?
6. In which year did Netflix add the highest number of titles?
7. Which countries produce the most Netflix content?
8. How does the average IMDb rating change over release years?
9. Which content type dominates Netflix: Movies or TV Shows?
10. Which sub-genres contribute most within major genres?

7. Selection of Appropriate Visuals

1. What is the total number of titles available on Netflix?

(Answered by KPI card: Total Titles)

2. What is the average IMDb rating of Netflix content?

(Answered by KPI card: Avg IMDb)

3. How many movies and TV shows are available on Netflix?

(Answered by Total Movies + Type donut chart)

4. Which genre has the highest number of titles?

(Answered by Main Genre bar chart)

5. How has Netflix content grown over the years?

(Answered by Cumulative Titles by Year chart)

6. In which year did Netflix add the highest number of titles?

(Answered by Release Year trend chart)

7. Which countries produce the most Netflix content?

(Answered by Production Countries map)

8. How does the average IMDb rating change over release years?

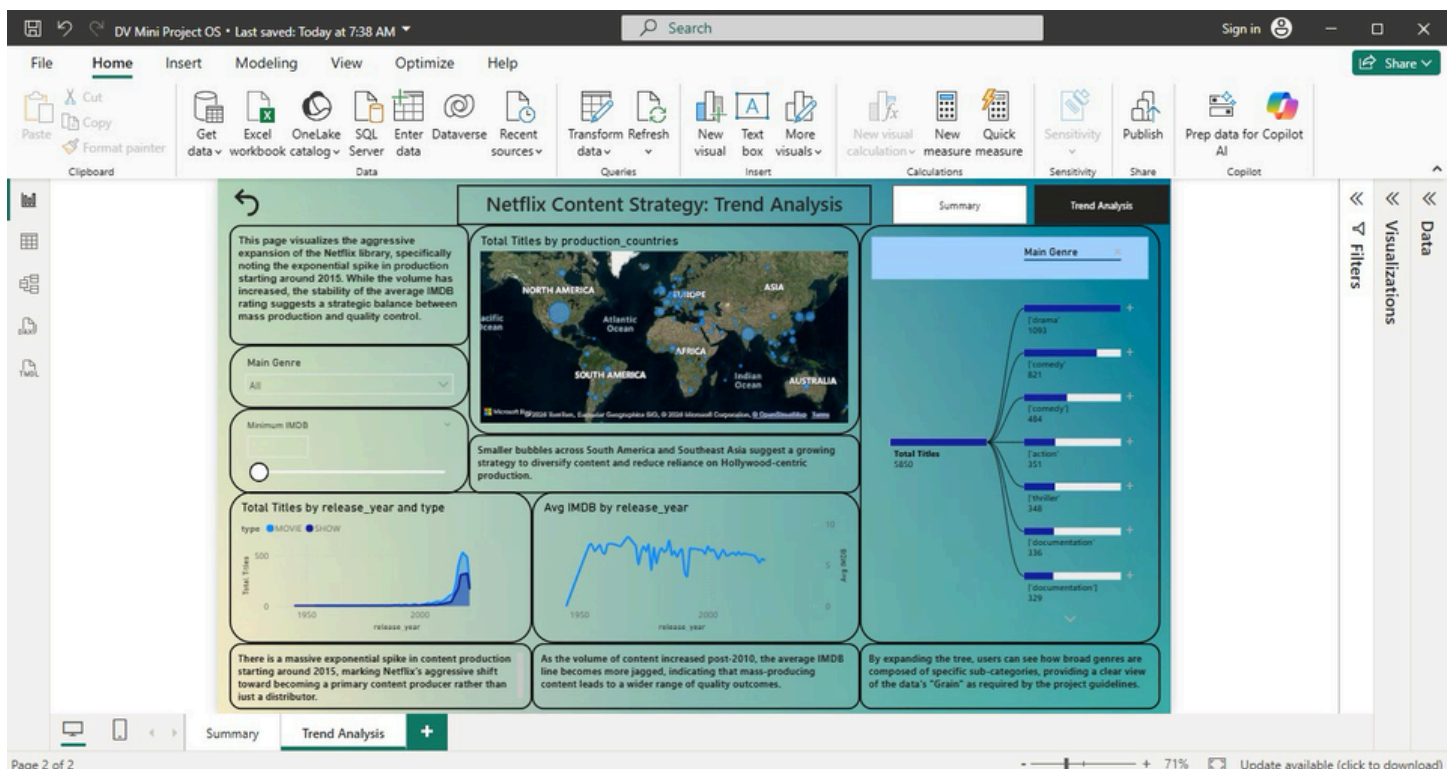
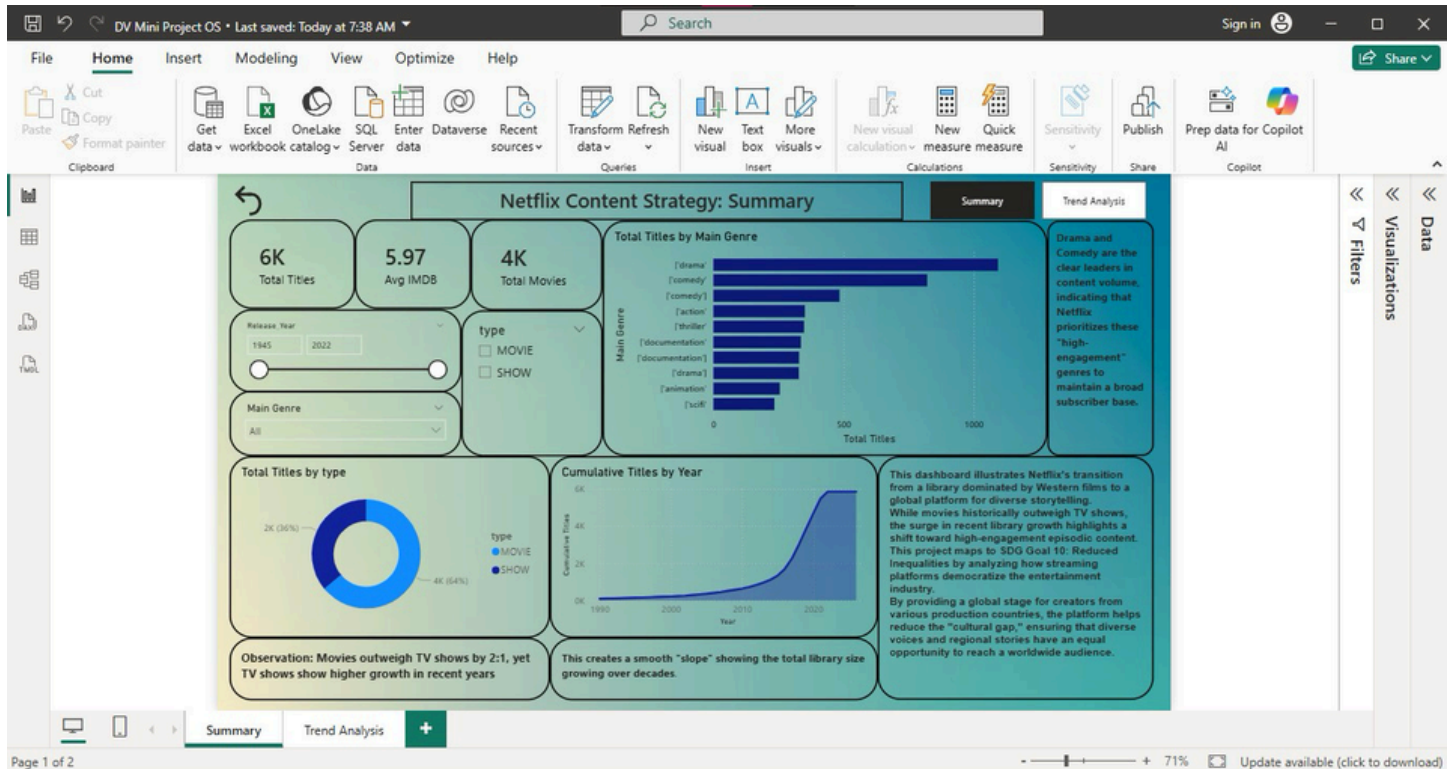
(Answered by Avg IMDb by Release Year line chart)

9. Which content type dominates Netflix: Movies or TV Shows?

(Answered by Type distribution donut chart)

10. Which sub-genres contribute most within major genres?

(Answered by Genre decomposition tree)



8. Create minimum 5 Measures

1. (DAX) : Total Titles = COUNT('Fact_titles'[id])

- **Measure Type: Measure** (Dynamic aggregation)
- This measure calculates the total number of Netflix content titles available in the dataset by counting all unique content IDs in the **Fact_titles** table. It helps provide an overall content volume and acts as a key KPI for the dashboard. The measure dynamically updates based on filters such as genre, release year, and content type.

The screenshot shows the Microsoft Power BI Desktop interface. At the top, the 'Measure tools' ribbon is active, displaying the 'Name' field as 'Total Titles', the 'Home table' as 'Fact_titles', and the 'Format' dropdown set to 'Whole number'. Below the ribbon, the 'Data' pane on the right shows a list of columns from the 'Fact_titles' table, including 'id', 'title', 'type', 'desc_length', 'description', 'genres', 'imdb_id', 'imdb_score', 'imdb_votes', 'is_highRated', 'mainGenre', 'production_countries', 'release_year', 'runtime', 'scoreRange', 'seasons', 'title', 'tmdb_popularity', 'tmdb_score', and 'Total Titles'. The 'Total Titles' measure is highlighted in the list. On the left, a preview of the data table is shown with columns 'id', 'title', and 'type'. The table contains 20 rows of data, including titles like 'Bill Hicks: Sane Man', 'Sam Kinison: Family Entertainment Hour', 'Blade Runner: The Final Cut', 'Apocalypse Now Redux', 'The Fighter', 'Jackass 2.5', 'Open Season 2', 'Doing Hard Time', 'Spookley the Square Pumpkin', 'To and from New York', 'Jackass 3.5', 'Hostel: Part III', 'BoJack Horseman Christmas Special', 'Dragons: Dawn Of The Dragon Racers', 'LEGO Marvel Super Heroes: Maximum Overload', 'Silent', 'The Meaning of Monty Python', 'From One Second to the Next', '100 Things to Do Before High School', and 'Todd Glass Talks About Stuff'.

id	title	type
tm132164	Bill Hicks: Sane Man	MOVIE
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE
tm7135	Bill Hicks: One Night Stand	MOVIE
tm997728	Blade Runner: The Final Cut	MOVIE
tm195620	Apocalypse Now Redux	MOVIE
tm101914	The Fighter	MOVIE
tm76428	Jackass 2.5	MOVIE
tm53999	Open Season 2	MOVIE
tm84348	Doing Hard Time	MOVIE
tm80487	Spookley the Square Pumpkin	MOVIE
tm404676	To and from New York	MOVIE
tm176507	Jackass 3.5	MOVIE
tm178395	Hostel: Part III	MOVIE
tm458034	BoJack Horseman Christmas Special	MOVIE
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE
tm137379	Silent	MOVIE
tm360904	The Meaning of Monty Python	MOVIE
tm182534	From One Second to the Next	MOVIE
tm132675	100 Things to Do Before High School	MOVIE
tm177376	Todd Glass Talks About Stuff	MOVIE

2. (DAX) : Avg IMDB = AVERAGE('Fact_titles'[imdb_score])

- **Measure Type: Measure** (Dynamic aggregation)
- This measure calculates the average IMDb rating of all Netflix titles in the dataset by averaging the values in the **imdb_score** column. It helps evaluate the overall quality of content available on Netflix and provides a benchmark for comparing genres, years, and content types. The measure dynamically updates based on applied filters.

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Avg IMDB = AVERAGE('Fact_titles'[imdb_score])

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It wa...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherd by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they have Wee Man use a 'pee' gu...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapp...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetra...
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyr...

Table: Fact_titles (5,850 rows) Column: Avg IMDB (0 distinct values)

Update available (click to download)

3. **(DAX) : Total Movies = CALCULATE([Total Titles], 'Fact_titles'[type] = "MOVIE")**

- **Measure Type: Measure** (Filtered aggregation)
- This measure calculates the total number of movies available in the Netflix dataset by filtering the content type as **MOVIE**. It helps distinguish movie content from TV shows and provides insights into Netflix's movie library size. The measure updates dynamically based on applied filters like genre, year, and rating.

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Total Movies = CALCULATE([Total Titles], 'Fact_titles'[type] = "MOVIE")

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It wa...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherd by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they have Wee Man use a 'pee' gu...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapp...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyr...

Table: Fact_titles (5,850 rows) Column: Total Movies (0 distinct values)

Update available (click to download)

4. **(DAX) : Total TV Shows = CALCULATE([Total Titles], 'Fact_titles'[type] = "SHOW")**
 - **Measure Type: Measure** (Filtered aggregation)
 - This measure calculates the total number of TV shows in the Netflix dataset by filtering the content type as **SHOW**. It helps analyze the volume of episodic content available on the platform and compare it with movies. The measure dynamically changes according to filters such as genre, release year, and ratings.

The screenshot shows the Microsoft Power BI Desktop interface. At the top, the title bar reads 'Untitled - Power BI Desktop'. Below it is a ribbon with tabs: 'File', 'Home', 'Help', 'Table tools', and 'Measure tools'. The 'Table tools' tab is active, showing options like 'Manage relationships', 'New relationships', 'Quick measure', 'New measure', 'New column', 'New table', and 'Mark as date table'. The 'Measure tools' tab is also visible. In the center, a DAX measure is defined: **1 Total TV Shows = CALCULATE([Total Titles], 'Fact_titles'[type] = "SHOW")**. Below the measure, a table of data is displayed with columns: 'id', 'title', and 'type'. The table contains 20 rows of movie data. On the right side, a 'Data' pane is visible, showing a search bar and a list of fields including 'description', 'genres', 'id', 'imdb_id', 'imdb_score', 'imdb_votes', 'Is High Rated', 'Main Genre', 'production_countries', 'release_year', 'runtime', 'Score Range', 'seasons', 'title', 'tmdb_popularity', 'tmdb_score', 'Total Movies', 'Total Titles', and 'Total TV Shows'. The 'Total TV Shows' measure is highlighted in the list.

id	title	type
tm132164	Bill Hicks: Sane Man	MOVIE
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE
tm7135	Bill Hicks: One Night Stand	MOVIE
tm997728	Blade Runner: The Final Cut	MOVIE
tm195620	Apocalypse Now Redux	MOVIE
tm101914	The Fighter	MOVIE
tm76428	Jackass 2.5	MOVIE
tm53999	Open Season 2	MOVIE
tm84348	Doing Hard Time	MOVIE
tm80487	Spokey the Square Pumpkin	MOVIE
tm404676	To and from New York	MOVIE
tm176507	Jackass 3.5	MOVIE
tm178395	Hostel: Part III	MOVIE
tm458034	BoJack Horseman Christmas Special	MOVIE
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE
tm137379	Silent	MOVIE
tm360904	The Meaning of Monty Python	MOVIE
tm182534	From One Second to the Next	MOVIE
tm132675	100 Things to Do Before High School	MOVIE
tm177376	Todd Glass Talks About Stuff	MOVIE

Table: Fact_titles (5,850 rows) Column: Total TV Shows (0 distinct values)

5. **(DAX) : High Quality % = DIVIDE(CALCULATE([Total Titles], 'Fact_titles'[imdb_score] > 8), [Total Titles],0)**
 - **Measure Type: Measure** (Percentage / Conditional aggregation)
 - This measure calculates the percentage of high-quality Netflix titles by identifying content with an **IMDb score greater than 8** and dividing it by the total number of titles. It helps measure the proportion of top-rated content available on the platform. The result updates dynamically based on filters such as genre, year, and content type.

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Manage relationships New measure Quick measure New table Mark as date table

Structure Relationships Calculations Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 High Quality % = DIVIDE(CALCULATE([Total Titles], 'Fact_titles'[imdb_score] > 8), [Total Titles], 0)

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It was...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherded by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what was left over from Jackass 2.0, and in this version they have Wee Man use a 'pee' gu...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapi...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spokey from the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyi...

Table: Fact_titles (5,850 rows) Column: High Quality % (0 distinct values)

Update available (click to download)

9. Create minimum 5 Calculated Columns

- (DAX) : Content Age = 2026 - 'Fact_titles'[release_year]**
 - Column Type: Calculated Column** (Derived column)
 - This calculated column determines the age of each Netflix title by subtracting its release year from the current year. It helps analyze how old or recent the content is in the Netflix library. This makes it useful for studying content freshness and release trends over time.

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Content Age

Data type Whole number

Format Whole number

Summarization Sum

Data category Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure Formatting Properties

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Content Age = 2026 - 'Fact_titles'[release_year]

genres	production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score	Content Age
comedy]	[US]	0	unknown	0	0	3.377	7.5	37
comedy]	[US]	0	unknown	0	0	1.84	6.1	35
comedy]	[US]	0	unknown	0	0	1.08	7.5	35
action', 'thriller', 'scifi']	[US]	0	unknown	0	0	0.055091	9	19
drama', 'war', 'horror', 'thriller']	[US]	0	unknown	0	0	4.147341	9.8	25
drama]	[US]	0	unknown	0	0	17.007	7.4	16
documentation', 'comedy']	[US]	0	unknown	0	0	28.197	6.1	19
comedy]	[US]	0	unknown	0	0	18.558	5.8	18
action', 'crime', 'drama']	[US]	0	unknown	0	0	7.867	5.3	22
animation]	[US]	0	unknown	0	0	1.552	4.6	22
	[US]	0	unknown	0	0	1.401	5.8	20
comedy', 'documentation', 'action']	[US]	0	unknown	0	0	19.565	6.3	15
horror]	[US]	0	unknown	0	0	32.7	5.1	15
	[US]	0	unknown	0	0	1.566	7.1	12
fantasy', 'animation', 'family', 'action', 'comedy']	[US]	0	unknown	0	0	41.761	7.2	12
action', 'animation', 'comedy']	[US]	0	unknown	0	0	10.788	6.2	13
animation', 'family', 'action']	[US]	0	unknown	0	0	3.596	7	12
documentation]	[US]	0	unknown	0	0	2.898	7.5	13
documentation]	[US]	0	unknown	0	0	4.02	7.1	13
	[US]	0	unknown	0	0	0.009441745879	0	12
comedy]	[US]	0	unknown	0	0	1.801	0	13

Table: Fact_titles (5,850 rows) Column: Content Age (63 distinct values)

Update available (click to download)

2. **(DAX) : Is High Rated = IF('Fact_titles'[imdb_score] >= 8.0, "High Rated", "Regular")**
 - **Column Type: Calculated Column** (Conditional classification)
 - This calculated column classifies Netflix titles based on their IMDb score into **High Rated** and **Regular** categories. Titles with an IMDb score of **8 or above** are marked as high-rated. This helps in filtering premium-quality content and analyzing top-performing titles separately.

Table: Fact_titles (5,850 rows) Column: Is High Rated (2 distinct values)

titles	production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score	Content Age	Is High Rated
...	[US]	0	unknown	0	0	3.377	7.5	37	Regular
...	[US]	0	unknown	0	0	1.84	6.1	35	Regular
...	[US]	0	unknown	0	0	1.08	7.5	35	Regular
...	[US]	0	unknown	0	0	0.055091	9	19	Regular
...	[US]	0	unknown	0	0	4.147341	9.8	25	Regular
...	[US]	0	unknown	0	0	17.007	7.4	16	Regular
...	[US]	0	unknown	0	0	28.197	6.1	19	Regular
...	[US]	0	unknown	0	0	18.558	5.8	18	Regular
...	[US]	0	unknown	0	0	7.867	5.3	22	Regular
...	[US]	0	unknown	0	0	1.552	4.6	22	Regular
...	[US]	0	unknown	0	0	1.401	5.8	20	Regular
...	[US]	0	unknown	0	0	19.565	6.3	15	Regular
...	[US]	0	unknown	0	0	32.7	5.1	15	Regular
...	[US]	0	unknown	0	0	1.566	7.1	12	Regular
...	[US]	0	unknown	0	0	41.761	7.2	12	Regular
...	[US]	0	unknown	0	0	10.788	6.2	13	Regular
...	[US]	0	unknown	0	0	3.596	7	12	Regular
...	[US]	0	unknown	0	0	2.898	7.5	13	Regular
...	[US]	0	unknown	0	0	4.02	7.1	13	Regular
...	[US]	0	unknown	0	0	0.009441745879	0	12	Regular
...	[US]	0	unknown	0	0	1.801	0	13	Regular

3. **(DAX) : Score Range = SWITCH(TRUE(), 'Fact_titles'[imdb_score] < 5, "Low", 'Fact_titles'[imdb_score] < 7.5, "Average", "High")**
 - **Column Type: Calculated Column** (Conditional classification)
 - This calculated column classifies Netflix titles based on their IMDb score into **High Rated** and **Regular** categories. Titles with an IMDb score of **8 or above** are marked as high-rated. This helps in filtering premium-quality content and analyzing top-performing titles separately.

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Fact_titles

Structure

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Score Range = SWITCH(TRUE(), 'Fact_titles'[imdb_score] < 5, "Low", 'Fact_titles'[imdb_score] < 7.5, "Average", "High")

production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score	Content Age	Is High Rated	Score Range
[US]	0	unknown	0	0	3.377	7.5	37	Regular	Low
[US]	0	unknown	0	0	1.84	6.1	35	Regular	Low
[US]	0	unknown	0	0	1.08	7.5	35	Regular	Low
[US]	0	unknown	0	0	0.055091	9	19	Regular	Low
[US]	0	unknown	0	0	4.147341	9.8	25	Regular	Low
[US]	0	unknown	0	0	17.007	7.4	16	Regular	Low
[US]	0	unknown	0	0	28.197	6.1	19	Regular	Low
[US]	0	unknown	0	0	18.558	5.8	18	Regular	Low
[US]	0	unknown	0	0	7.867	5.3	22	Regular	Low
[US]	0	unknown	0	0	1.552	4.6	22	Regular	Low
[US]	0	unknown	0	0	1.401	5.8	20	Regular	Low
[US]	0	unknown	0	0	19.565	6.3	15	Regular	Low
[US]	0	unknown	0	0	32.7	5.1	15	Regular	Low
[US]	0	unknown	0	0	1.566	7.1	12	Regular	Low
[US]	0	unknown	0	0	41.761	7.2	12	Regular	Low
[US]	0	unknown	0	0	10.788	6.2	13	Regular	Low
[US]	0	unknown	0	0	3.596	7	12	Regular	Low
[US]	0	unknown	0	0	2.898	7.5	13	Regular	Low
[US]	0	unknown	0	0	4.02	7.1	13	Regular	Low
[US]	0	unknown	0	0	0.009441745879	0	12	Regular	Low
[US]	0	unknown	0	0	1.801	0	13	Regular	Low

Table: Fact_titles (5,850 rows) Column: Score Range (3 distinct values)

Update available (click to download)

4. (DAX) : Desc Length = LEN('Fact_titles'[description])

- Column Type: Calculated Column (Text length derivation)
- This calculated column calculates the total number of characters in each Netflix title's description. It helps measure the length and richness of content metadata. This can be useful for analyzing how detailed content descriptions are across different genres and content types.

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Fact_titles

Structure

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Desc Length = LEN('Fact_titles'[description])

production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score	Content Age	Is High Rated	Score Range	Desc Length
0	unknown	0	0	0	3.377	7.5	37	Regular	Low	313
0	unknown	0	0	0	1.84	6.1	35	Regular	Low	491
0	unknown	0	0	0	1.08	7.5	35	Regular	Low	130
0	unknown	0	0	0	0.055091	9	19	Regular	Low	657
0	unknown	0	0	0	4.147341	9.8	25	Regular	Low	338
0	unknown	0	0	0	17.007	7.4	16	Regular	Low	226
0	unknown	0	0	0	28.197	6.1	19	Regular	Low	326
0	unknown	0	0	0	18.558	5.8	18	Regular	Low	198
0	unknown	0	0	0	7.867	5.3	22	Regular	Low	150
0	unknown	0	0	0	1.552	4.6	22	Regular	Low	969
0	unknown	0	0	0	1.401	5.8	20	Regular	Low	326
0	unknown	0	0	0	19.565	6.3	15	Regular	Low	209
0	unknown	0	0	0	32.7	5.1	15	Regular	Low	152
0	unknown	0	0	0	1.566	7.1	12	Regular	Low	145
0	unknown	0	0	0	41.761	7.2	12	Regular	Low	143
0	unknown	0	0	0	10.788	6.2	13	Regular	Low	144
0	unknown	0	0	0	3.596	7	12	Regular	Low	107
0	unknown	0	0	0	2.898	7.5	13	Regular	Low	126
0	unknown	0	0	0	4.02	7.1	13	Regular	Low	118
0	unknown	0	0	0	0.009441745879	0	12	Regular	Low	186
0	unknown	0	0	0	1.801	0	13	Regular	Low	483

Table: Fact_titles (5,850 rows) Column: Desc Length (716 distinct values)

Update available (click to download)

5. **(DAX) : Main Genre = LEFT('Fact_titles'[genres], SEARCH(";", 'Fact_titles'[genres] & ";", 1) - 1)**
 - **Column Type: Calculated Column** (Text extraction)
 - This calculated column extracts the primary or first-listed genre from the genres field for each Netflix title. It simplifies genre analysis by converting multiple genre values into a single main genre. This makes filtering, grouping, and visualization more efficient in the dashboard.

Table: Fact_titles (5,850 rows) Column: Main Genre (35 distinct values)

10. Apply minimum 7 Time Intelligence Functions

1. **(DAX) : Titles YTD =TOTALYTD([Total Titles],'Calendar'[Date])**
 - **Measure Type: Time Intelligence Measure** (Year-to-Date aggregation)
 - This measure calculates the cumulative total number of Netflix titles from the beginning of the year up to the selected date. It helps track yearly content growth and analyze how Netflix's content library expands over time. The measure updates dynamically based on date filters.

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Titles YTD = TOTALYTD([Total Titles], 'Calendar'[Date])

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It was...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherd by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they have Wee Man use a 'pee' gu...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapp...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyi...

Table: Fact_titles (5,850 rows) Column: Titles YTD (0 distinct values)

Update available (click to download)

2. **(DAX) : Titles LY =CALCULATE([Total Titles],SAMEPERIODLASTYEAR('Calendar'[Date]))**
 - o **Measure Type: Time Intelligence Measure** (Last Year comparison)
 - o This measure calculates the total number of Netflix titles from the same period in the previous year. It helps perform year-over-year comparisons and analyze content growth trends across different years. The measure dynamically changes according to the selected date context

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Titles LY = CALCULATE([Total Titles], SAMEPERIODLASTYEAR('Calendar'[Date]))

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It was...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherd by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they have Wee Man use a 'pee' gu...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapp...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyi...

Table: Fact_titles (5,850 rows) Column: Titles LY (0 distinct values)

Update available (click to download)

3. **(DAX) : YoY Growth = [Total Titles] - [Titles LY]**
 - o **Measure Type: Time Intelligence Measure** (Year-over-Year growth)

- This measure calculates the year-over-year growth in Netflix titles by subtracting the previous year's total titles from the current total titles. It helps identify whether content production has increased or decreased compared to last year. This measure is useful for growth trend analysis.

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Measure' button is selected. The 'Name' field is set to 'Fact_titles'. The 'Measure' field is set to '1 YoY Growth = [Total Titles] - [Titles LY]'. The 'Data' pane on the right shows the 'Fact_titles' table with columns: id, title, type, description, genres, High Quality %, imdb_id, imdb_score, imdb_votes, Is High Rated, Main Genre, production_countries, release_year, runtime, Score Range, seasons, title, Titles LY, Titles YTD, tmdb_popularity, and tmdb_score. The table has 5,850 rows and 0 distinct values for the YoY Growth column.

4. (DAX) : YoY Growth % =DIVIDE([YoY Growth], [Titles LY], 0)

- **Measure Type: Time Intelligence Measure** (Year-over-Year growth percentage)
- This measure calculates the percentage growth in Netflix titles compared to the same period in the previous year. It divides the year-over-year growth by last year's total titles to show the growth rate. This helps analyze how fast Netflix's content library is expanding over time

The screenshot shows the Power BI Desktop interface. The 'Table tools' ribbon is active, and the 'Measure' button is selected. The 'Name' field is set to 'Fact_titles'. The 'Measure' field is set to '1 YoY Growth % = DIVIDE([YoY Growth], [Titles LY], 0)'. The 'Data' pane on the right shows the 'Fact_titles' table with columns: id, title, type, description, genres, High Quality %, imdb_id, imdb_score, imdb_votes, Is High Rated, Main Genre, production_countries, release_year, runtime, Score Range, seasons, title, Titles LY, Titles YTD, tmdb_popularity, and tmdb_score. The table has 5,850 rows and 0 distinct values for the YoY Growth % column.

5. **(DAX) : Rolling 3Yr Avg = AVERAGEX(DATESINPERIOD('Calendar'[Date], LASTDATE('Calendar'[Date]), -3, YEAR), [Total Titles])**
 - **Measure Type: Time Intelligence Measure** (Rolling average)
 - This measure calculates the rolling average of Netflix titles over the last **3 years** based on the selected date context. It helps smooth out short-term fluctuations and identify long-term content growth trends. This is useful for analyzing consistency in Netflix's content expansion strategy.

The screenshot shows the Power BI Desktop interface. The top ribbon includes 'Table tools' and 'Measure tools'. The 'Name' field is set to 'Fact_titles'. The 'Measure' field contains the DAX formula: `1 Rolling 3Yr Avg = AVERAGEX(DATESINPERIOD('Calendar'[Date], LASTDATE('Calendar'[Date]), -3, YEAR), [Total Titles])`. The table below lists various Netflix titles with their IDs, titles, and types. The right sidebar shows the 'Data' pane with a search bar and a list of fields including 'description', 'genres', 'High Quality %', 'id', 'imdb_id', 'imdb_score', 'imdb_votes', 'Is High Rated', 'Main Genre', 'production_countries', 'release_year', 'Rolling 3Yr Avg', 'runtime', 'Score Range', 'seasons', 'title', 'Titles LY', 'Titles YTD', and 'tmdb_popularity'.

id	title	type
tm132164	Bill Hicks: Sane Man	MOVIE
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE
tm7135	Bill Hicks: One Night Stand	MOVIE
tm997728	Blade Runner: The Final Cut	MOVIE
tm195620	Apocalypse Now Redux	MOVIE
tm101914	The Fighter	MOVIE
tm76428	Jackass 2.5	MOVIE
tm53999	Open Season 2	MOVIE
tm84348	Doing Hard Time	MOVIE
tm80487	Spookley the Square Pumpkin	MOVIE
tm404676	To and from New York	MOVIE
tm176507	Jackass 3.5	MOVIE
tm178395	Hostel: Part III	MOVIE
tm458034	Bolack Horseman Christmas Special	MOVIE
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE
tm137379	Silent	MOVIE
tm360904	The Meaning of Monty Python	MOVIE
tm182534	From One Second to the Next	MOVIE
tm132675	100 Things to Do Before High School	MOVIE
tm177376	Todd Glass Talks About Stuff	MOVIE

6. **(DAX) : Cumulative Titles = CALCULATE([Total Titles], FILTER(ALL('Calendar'), 'Calendar'[Date] <= MAX('Calendar'[Date])))**
 - **Measure Type: Time Intelligence Measure** (Running total / Cumulative aggregation)
 - This measure calculates the cumulative total number of Netflix titles over time by continuously adding titles up to the selected date. It helps visualize overall content growth and expansion trends. This measure is useful for understanding Netflix's long-term content accumulation strategy.

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Cumulative Titles = CALCULATE([Total Titles], FILTER(ALL('Calendar'), 'Calendar'[Date] <= MAX('Calendar'[Date])))

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It wa...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they h...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight detour when Mr. Weenie is kidnapp...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyr...

Table: Fact_titles (5,850 rows) Column: Cumulative Titles (0 distinct values)

Update available (click to download)

Data

Search

- Dim_credits
 - character
 - id
 - name
 - person_id
 - role
- Fact_titles
 - age_certification
 - Avg IMDB
 - Content Age
 - Cumulative Titles
 - Desc Length
 - description
 - genres
 - High Quality %
 - id
 - imdb_id
 - imdb_score
 - imdb_votes

7. **(DAX) : Titles PQ = CALCULATE([Total Titles], PREVIOUSQUARTER('Calendar'[Date]))**
- Measure Type: Time Intelligence Measure (Previous Quarter comparison)
 - This measure calculates the total number of Netflix titles from the **previous quarter** based on the selected date context. It helps compare quarterly content performance and identify short-term growth trends. This is useful for analyzing Netflix's content expansion quarter by quarter

Untitled - Power BI Desktop

File Home Help Table tools Measure tools

Name Fact_titles

Structure Relationships Calculations Mark as date table Calendars

Auto recovery contains some recovered files that haven't been opened. View recovered files

1 Titles PQ = CALCULATE([Total Titles], PREVIOUSQUARTER('Calendar'[Date]))

id	title	type	
tm132164	Bill Hicks: Sane Man	MOVIE	Sane Man was filmed before Bill recorded 'Dangerous', his first comedy album, and is a turning point in Hicks' career. It wa...
tm8687	Sam Kinison: Family Entertainment Hour	MOVIE	Get ready to scream with laughter! Sam Kinison's primal scream will echo down through the comedy hall of fame summing...
tm7135	Bill Hicks: One Night Stand	MOVIE	Bill Hicks shows us his view on smoking, smoking pot, drinking, sex, advertisement and music. It's not a view you hear very...
tm997728	Blade Runner: The Final Cut	MOVIE	Visually spectacular, intensely action-packed and powerfully prophetic since its debut, Blade Runner returns in Ridley Scott's...
tm195620	Apocalypse Now Redux	MOVIE	Apocalypse Now Redux is a 2001 extended version of Francis Ford Coppola's epic war film Apocalypse Now, which was origi...
tm101914	The Fighter	MOVIE	Boxer "Irish" Micky Ward's unlikely road to the world light welterweight title. His Rocky-like rise was shepherd by half-br...
tm76428	Jackass 2.5	MOVIE	The crew have now set off to finish what as left over from Jackass 2.0, and in this version they h...
tm53999	Open Season 2	MOVIE	After falling head over hooves in love with Giselle, Elliot's road to the altar takes a slight deto...
tm84348	Doing Hard Time	MOVIE	A father's eight year old son is murdered in a gangland shooting. The father (Boris Kodjoe) gets himself thrown into prison...
tm80487	Spookley the Square Pumpkin	MOVIE	Dare to be square! In a world where "The only good pumpkins are round pumpkins!" SPOOKLEY THE SQUARE PUMPKIN is...
tm404676	To and from New York	MOVIE	While covering a story in New York City, a Seattle-based reporter uncovers a link between two missing women that change...
tm176507	Jackass 3.5	MOVIE	Johnny Knoxville of 'Jackass' releases unused material of stunts, tricks, antics and shenanigans shot during the production c...
tm178395	Hostel: Part III	MOVIE	Set in Las Vegas, the film centers on a man who attends his best friend's bachelor party, unaware of an insidious agenda th...
tm458034	BoJack Horseman Christmas Special	MOVIE	It's Christmas, and BoJack wants nothing to do with it. Then Todd shows up with a giant candy cane and an old "Horsin' Arc...
tm139617	Dragons: Dawn Of The Dragon Racers	MOVIE	A hunt for a lost sheep turns into a competition between Hiccup and friends as they compete to become the first Dragon F...
tm180802	LEGO Marvel Super Heroes: Maximum Overload	MOVIE	Villainous Loki is amassing an army to conquer Earth! His antics are keeping Spider-Man and S.H.I.E.L.D. busy as they tackle...
tm137379	Silent	MOVIE	A duo of street performers learns how sound and picture work together to create amazing cinema experiences.
tm360904	The Meaning of Monty Python	MOVIE	A discussion between John Cleese, Michael Palin, Eric Idle, Terry Gilliam and Terry Jones about their film The Meaning of Li...
tm182534	From One Second to the Next	MOVIE	Stories of serious traffic accidents caused by texting and driving are told by the perpetrators and surviving victims.
tm132675	100 Things to Do Before High School	MOVIE	With the pressures of high school in their near future, CJ and her two best friends decide to embark on a series of middle s...
tm177376	Todd Glass Talks About Stuff	MOVIE	In Todd Glass Talks About Stuff, Todd Glass talks about...stuff, covering infomercials, ranting about eating slowly, and bullyr...

Table: Fact_titles (5,850 rows) Column: Titles PQ (0 distinct values)

Update available (click to download)

Data

Search

- Main Genre
- production_countries
- release_year
- Rolling 3Yr Avg
- runtime
- Score Range
- seasons
- title
- Titles LY
- Titles PQ
- Titles YTD
- tmdb_popularity
- tmdb_score
- Total Movies
- Total Titles
- Total TV Shows
- type
- YoY Growth
- YoY Growth %

11. Publish Report to Power BI Service

- After completing the dashboard development and analysis in Power BI Desktop, the final report is published to the Microsoft Power BI Service for cloud-based access and sharing. Publishing the report allows team members, faculty, and stakeholders to view the dashboard online, interact with visuals, and access real-time insights from any device. This step improves collaboration, enables report sharing through web links, and ensures easy accessibility of the project for presentation and evaluation purposes.

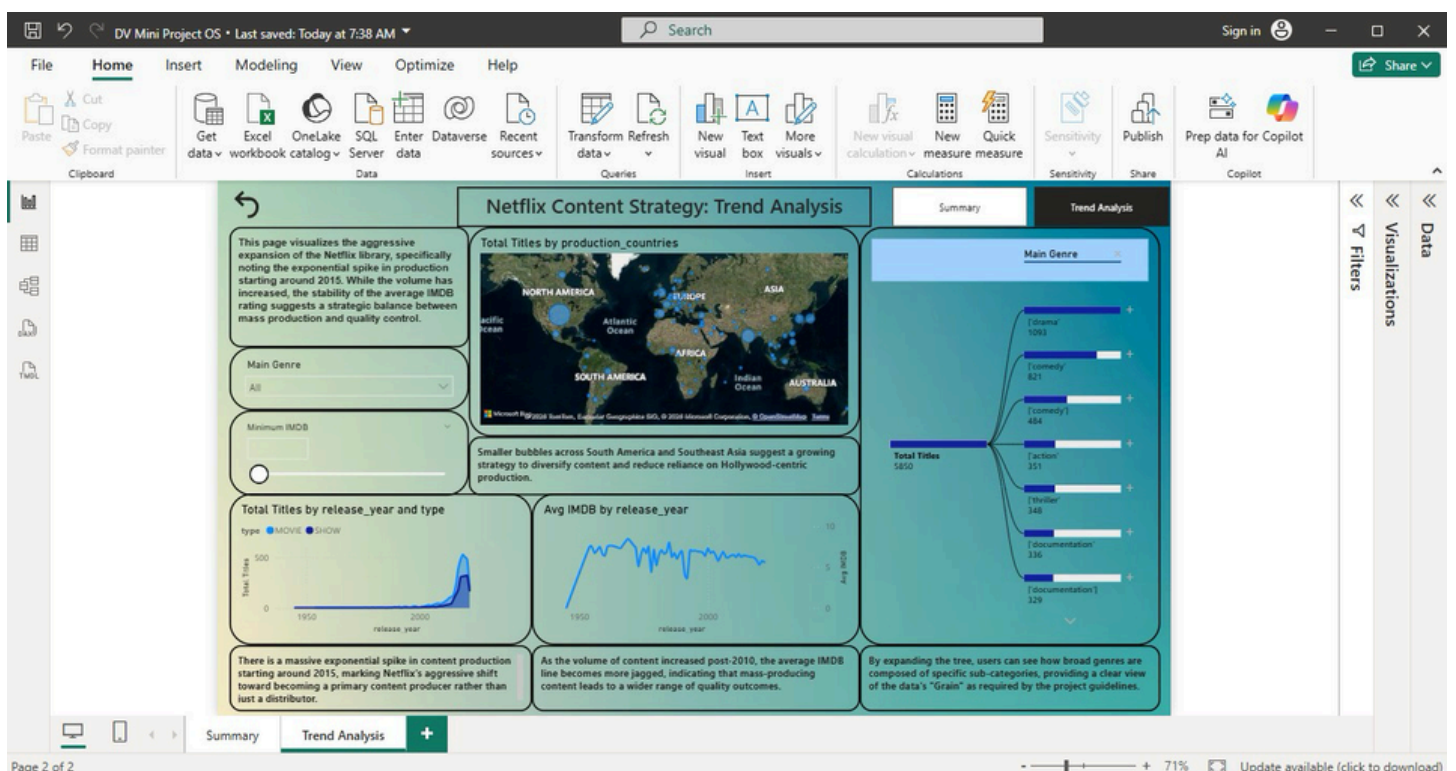
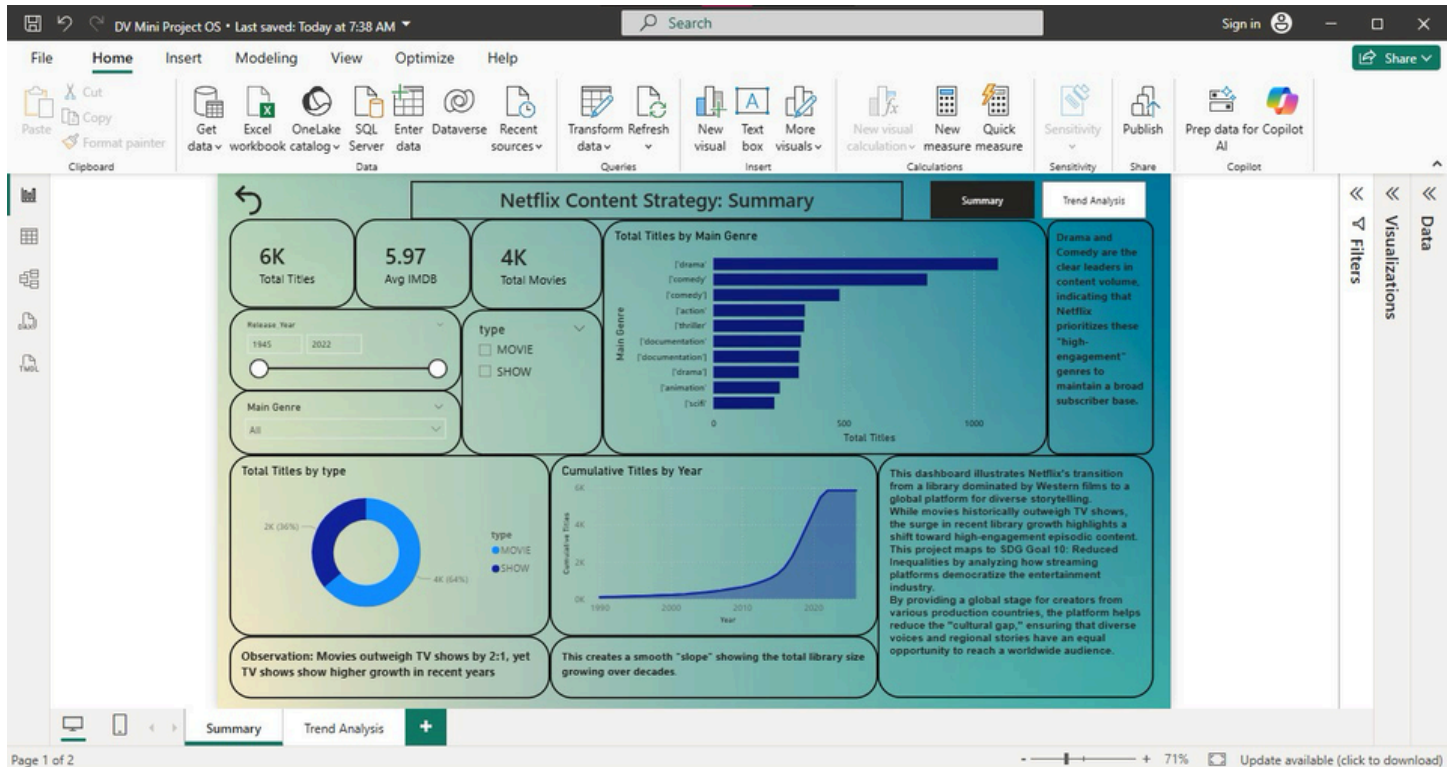
12. Map your project to a relevant SDG Goal

- This project is mapped to United Nations **SDG Goal 10: Reduced Inequalities** because streaming platforms like Netflix provide equal access to entertainment content across different regions, cultures, and communities. By analyzing Netflix's global content distribution, genre diversity, and production countries, the project highlights how digital platforms help reduce cultural and regional content gaps. This supports inclusivity by giving creators from different countries a platform to reach a worldwide audience and making diverse stories accessible to everyone.

13. Provide meaningful Insights / Inferences for each visual

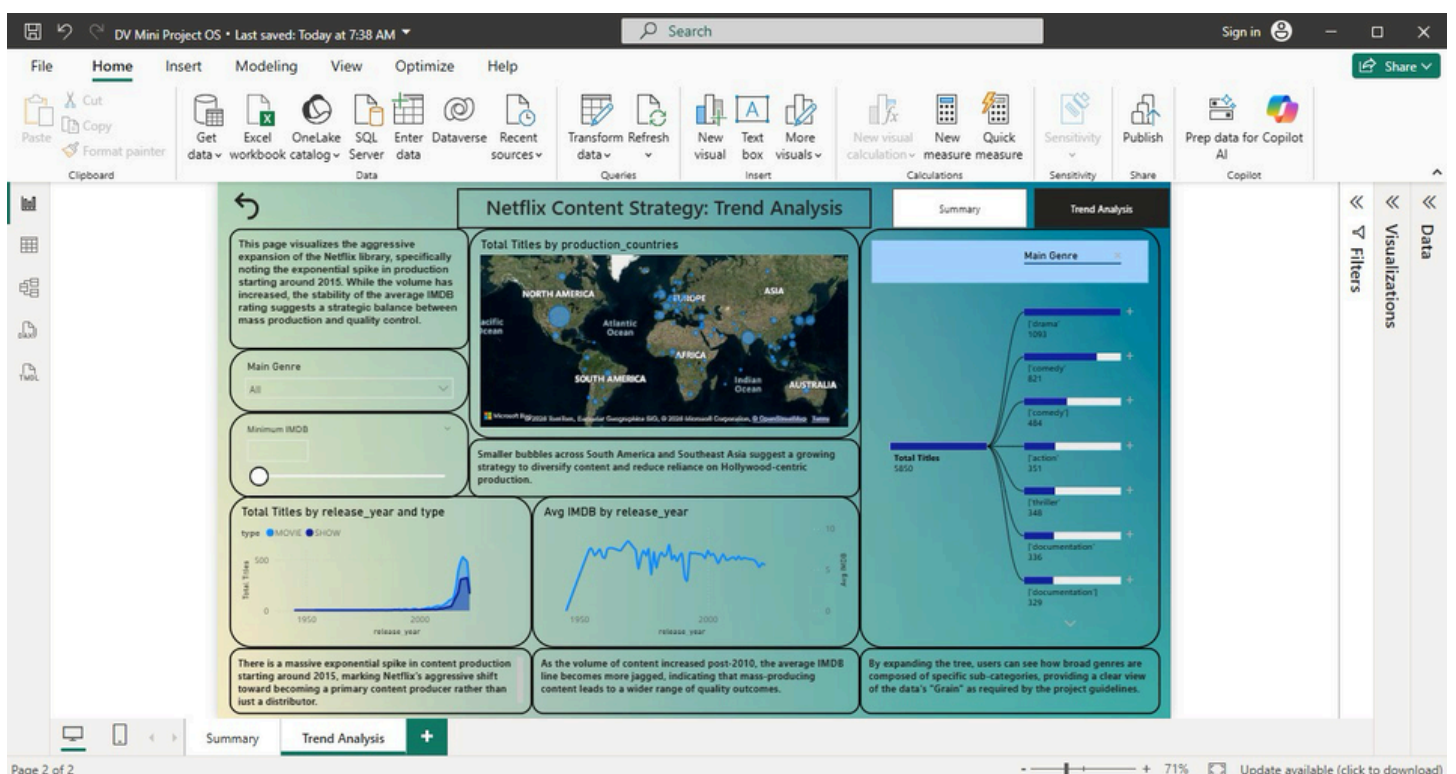
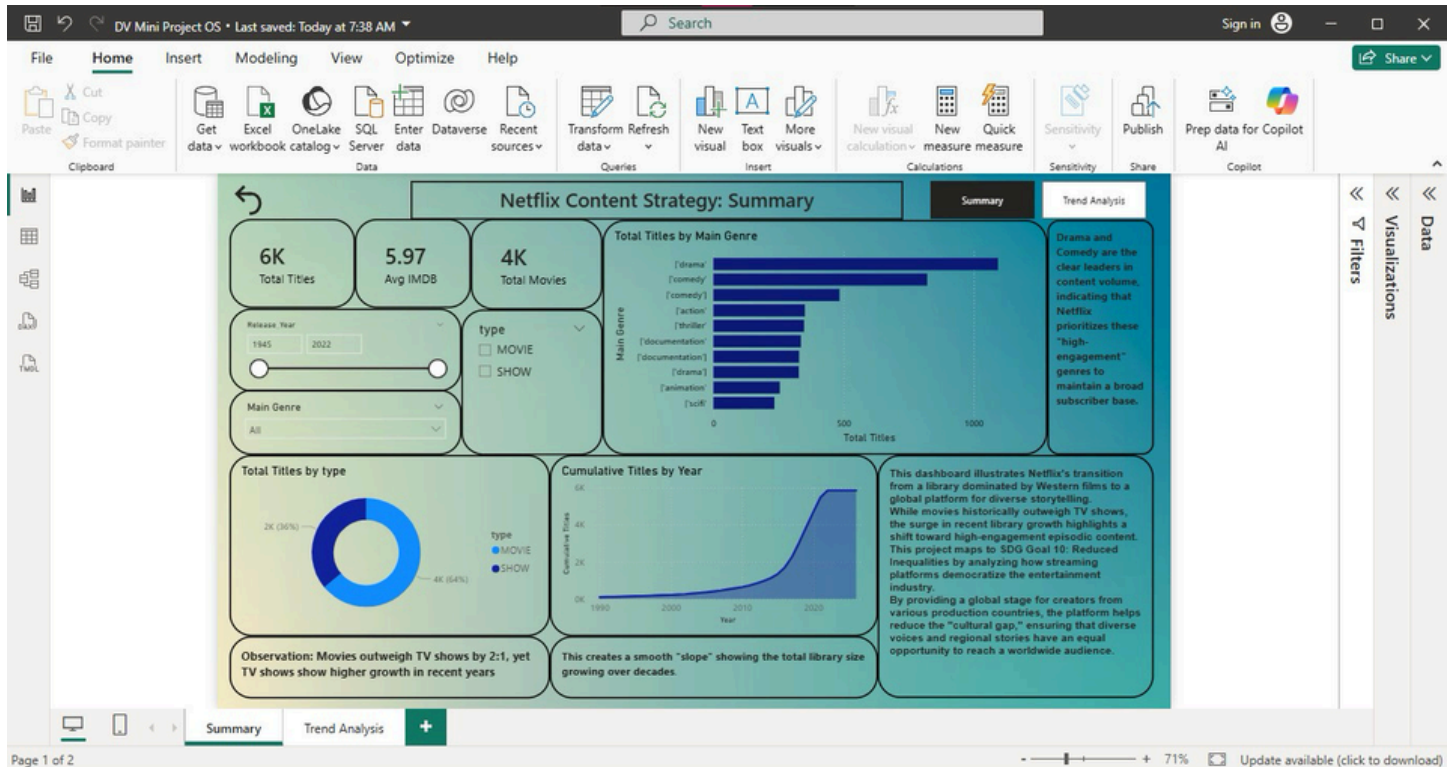
- **Total Titles (KPI Card)**
 - Netflix has a content library of approximately **6,000 titles**.
 - The platform maintains a large and diverse content collection.
 - High content volume indicates strong platform engagement strategy.
- **Average IMDb (KPI Card)**
 - The average IMDb rating is around **5.97**, indicating moderate content quality.
 - Netflix maintains a balance between content quantity and quality.
 - Ratings show a wide mix of premium and regular content.
- **Total Movies (KPI Card)**
 - Movies dominate the platform with around **4,000 titles**.
 - Netflix's content strategy historically focuses more on movies.
 - Movie content remains the primary content type.
- **Total Titles by Main Genre (Bar Chart)**
 - **Drama** is the most dominant genre.
 - Comedy is the second most preferred genre.
 - Netflix prioritizes high-engagement genres for mass audience appeal.
- **Total Titles by Type (Donut Chart)**
 - Movies contribute approximately **64%** of total content.
 - TV Shows contribute around **36%**.
 - TV Shows are growing but Movies remain dominant.
- **Cumulative Titles by Year (Area Chart)**
 - Major content growth starts after **2015**.
 - Netflix shifted from content distribution to content production.
 - The content library expanded rapidly in the last decade.
- **Production Countries Map (Map Visual)**
 - North America has the highest content production.
 - Europe and Asia are growing contributors.

- Netflix is expanding its regional content strategy globally.
- **Total Titles by Release Year and Type (Line Chart)**
 - Content production increased sharply after 2015.
 - TV Shows show stronger recent growth than movies.
 - Netflix has increased original content production significantly.
- **Avg IMDb by Release Year (Line Chart)**
 - Average IMDb remains relatively stable over time.
 - Higher content volume has increased rating variation.
 - Netflix maintains quality consistency despite mass production.
- **Genre Decomposition Tree**
 - Drama contributes the highest number of titles.
 - Genre hierarchy shows strong content diversification.
 - Sub-genres reveal Netflix's deeper content segmentation strategy.



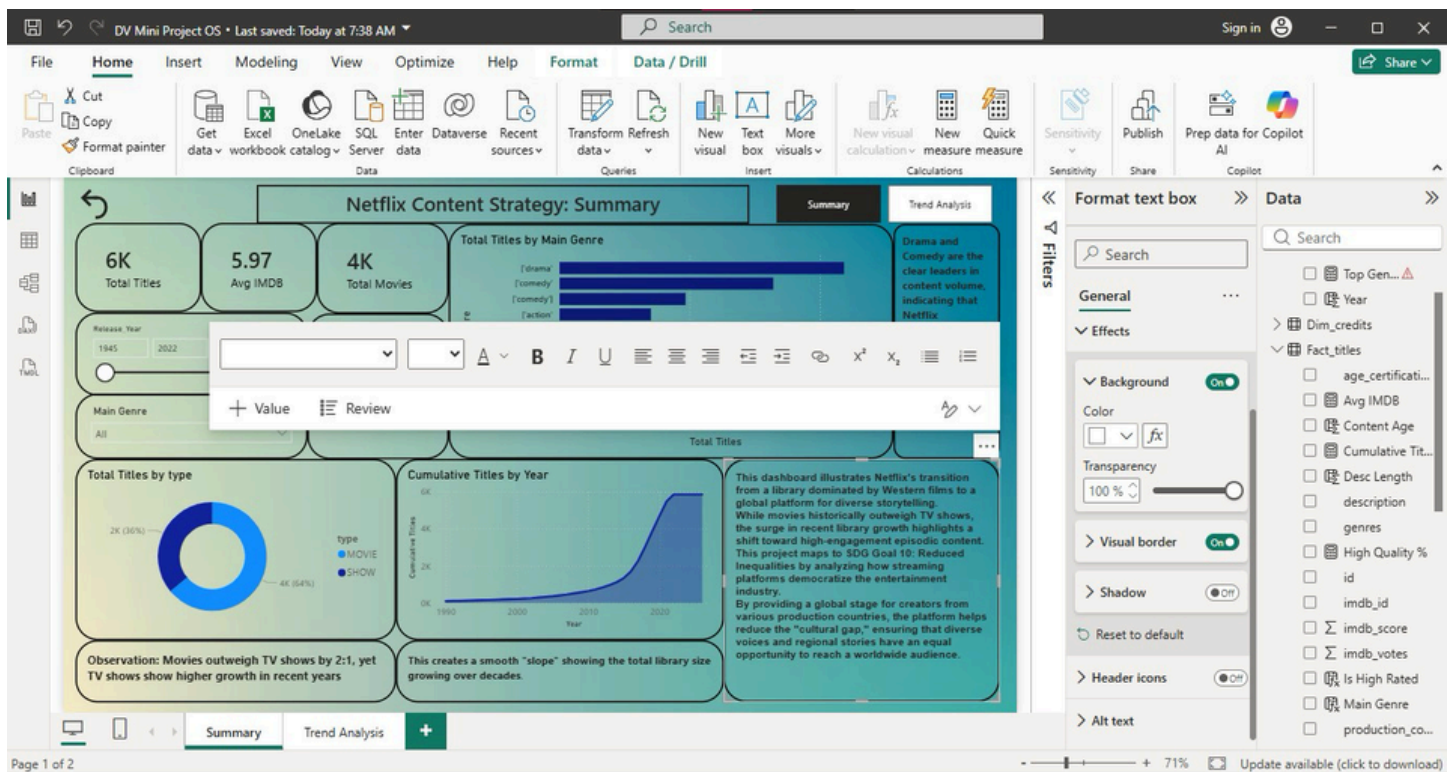
14. Create a proper Dashboard / Storyboard

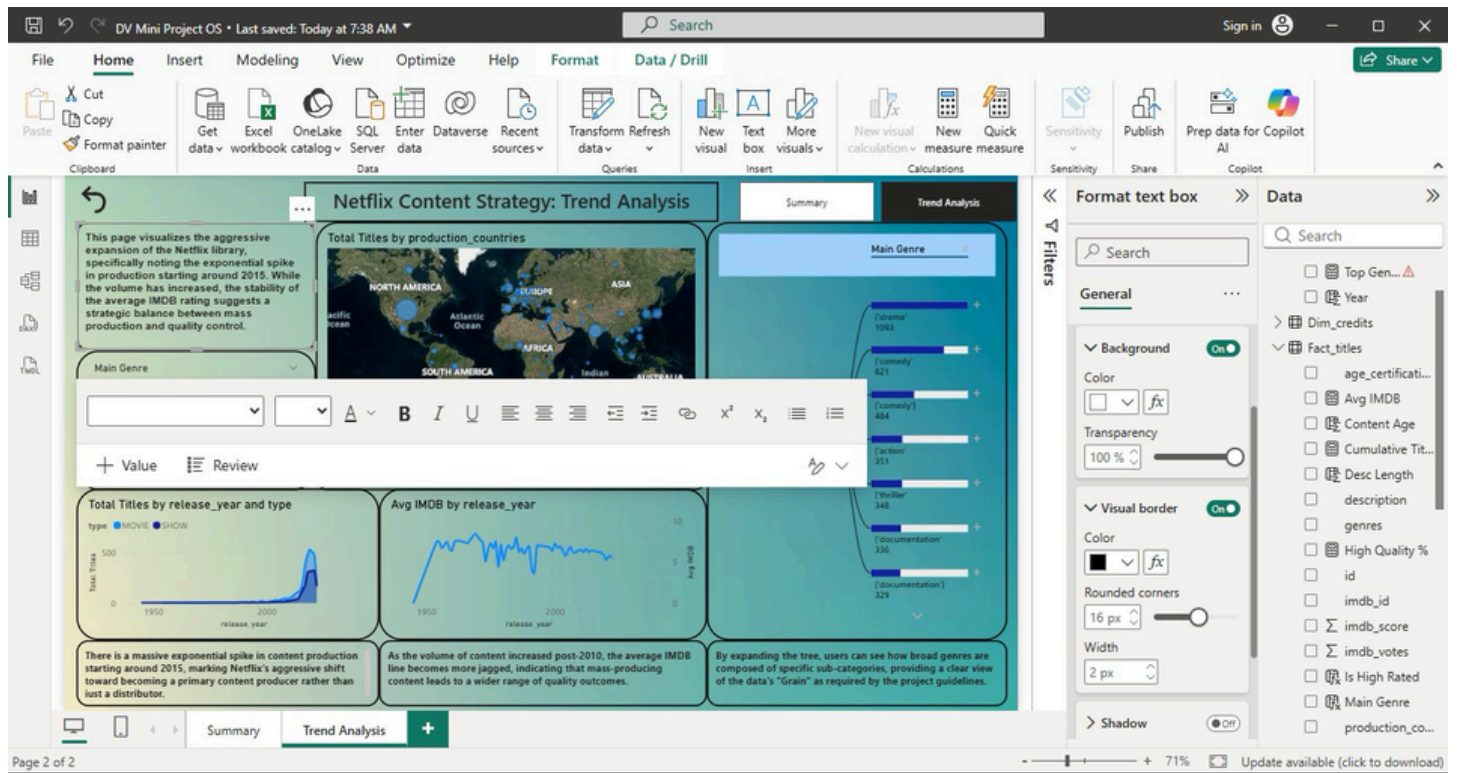
- The dashboard is designed in a structured storyboard format to present Netflix content analysis in a clear and logical flow. It is divided into two main pages: Summary Dashboard and Trend Analysis Dashboard. The Summary page provides high-level insights such as total titles, average IMDb ratings, content type distribution, genre analysis, and cumulative growth trends, while the Trend Analysis page focuses on production country distribution, release year trends, quality trends, and genre decomposition. The dashboard follows a storytelling approach where users can move from overview to detailed analysis using page navigation, slicers, and interactive visuals, making the report organized, user-friendly, and easy to interpret.



15. Apply Data Storytelling techniques

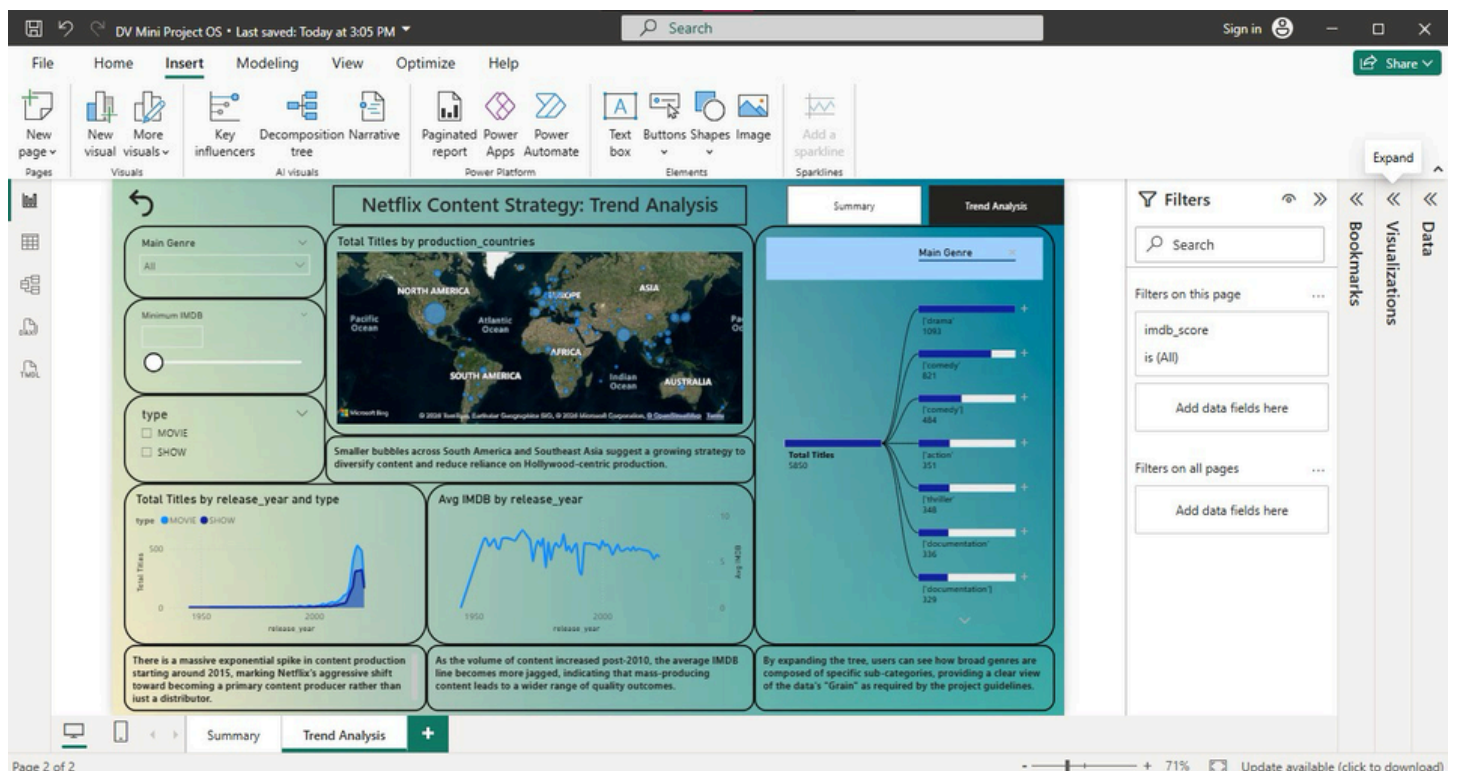
- Data storytelling techniques are applied in this project by combining visuals, KPIs, filters, and insight panels to create a clear narrative about Netflix's content strategy. The dashboard starts with high-level summary metrics such as total titles, average IMDb ratings, and content type distribution, then gradually moves into deeper trend analysis like genre dominance, yearly growth, production country distribution, and content quality trends. Insight boxes are added alongside visuals to explain important observations and business meanings, helping users understand not just the data, but the story behind Netflix's expansion, content diversity, and evolving production strategy.

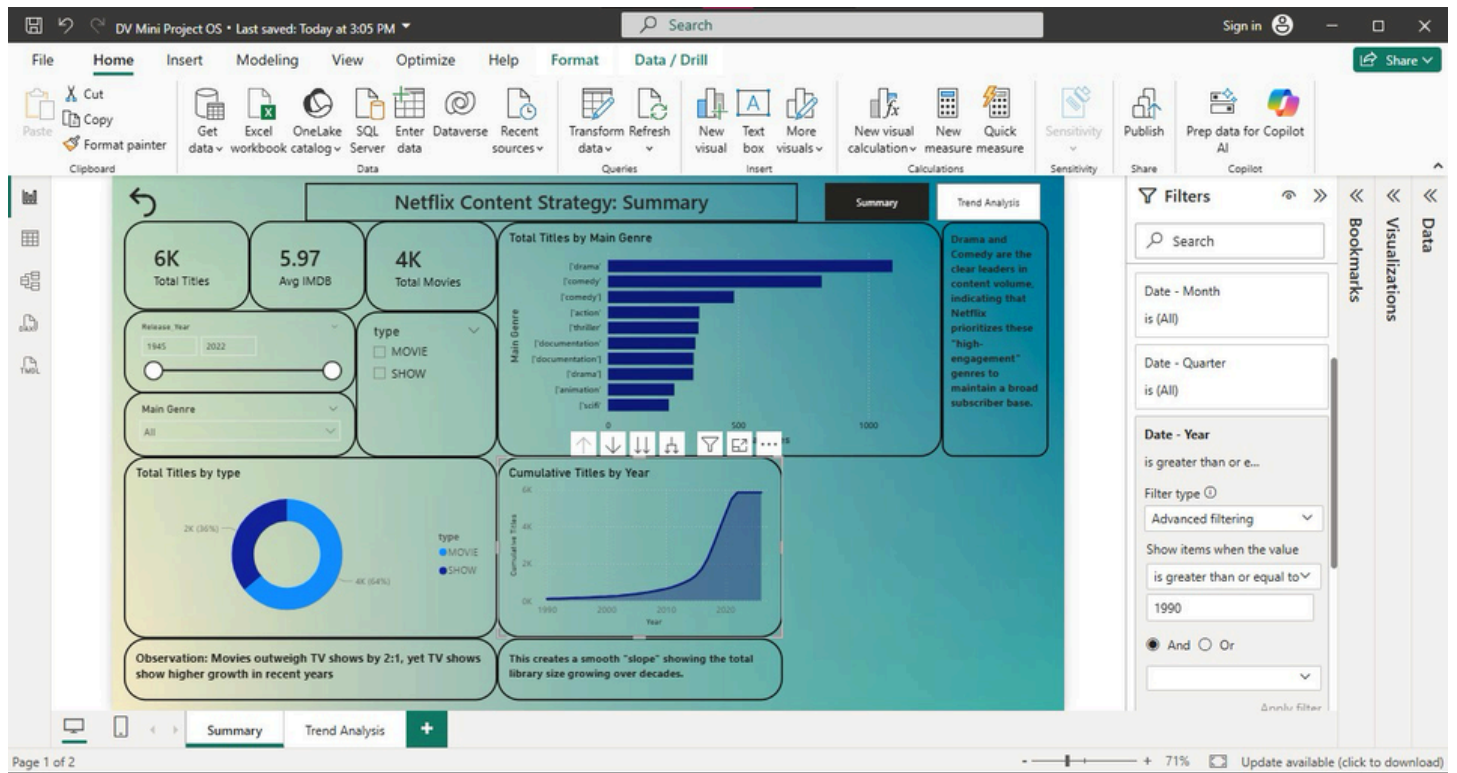




16. Use Slicers and Filters effectively

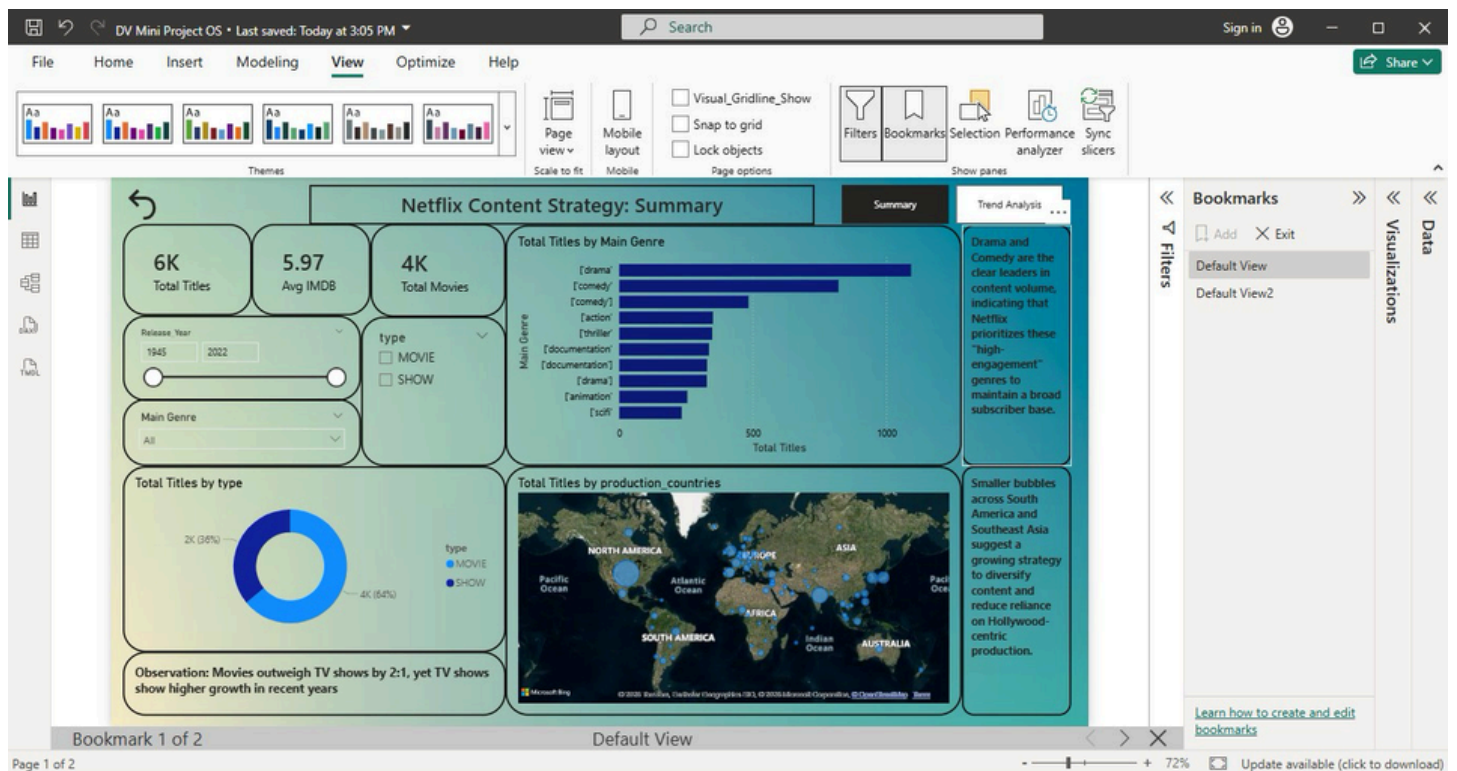
- Slicers and filters are effectively used in this dashboard to make the analysis interactive and user-friendly. Slicers such as **Release Year**, **Content Type**, **Main Genre**, and **Minimum IMDb Score** allow users to filter the Netflix dataset dynamically and explore specific content categories based on their interests. These slicers are connected to all visuals on the dashboard, enabling real-time cross-filtering and focused analysis. This improves user experience by making the dashboard flexible, interactive, and capable of providing customized insights.





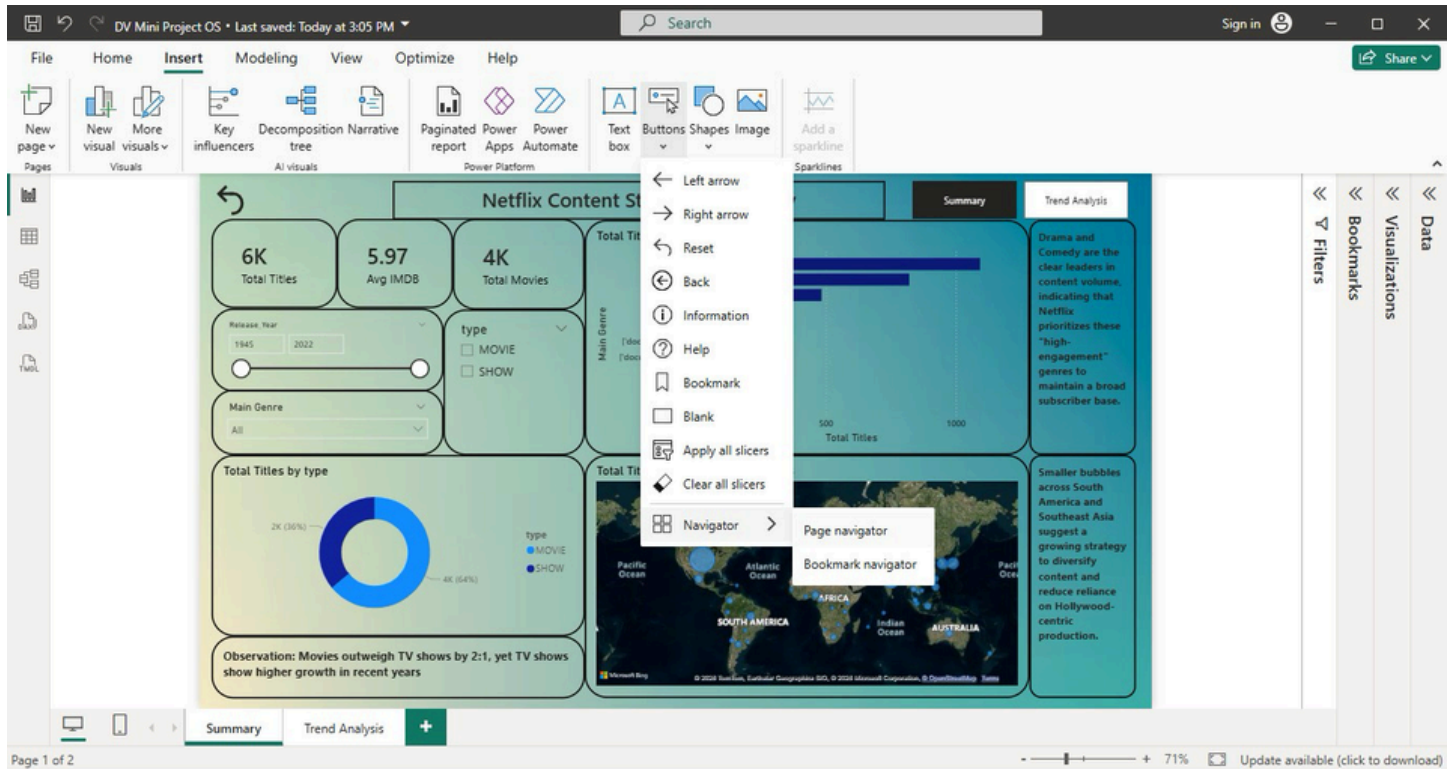
17. Implement Bookmarks

- Bookmarks are implemented in the Power BI dashboard to save specific report views and improve user navigation. They allow users to quickly switch between predefined dashboard states, such as Summary View, Trend Analysis View, or Genre-Specific Analysis, without manually applying filters each time. This enhances interactivity, supports data storytelling, and provides a smoother user experience by enabling quick access to important insights and different report perspectives.



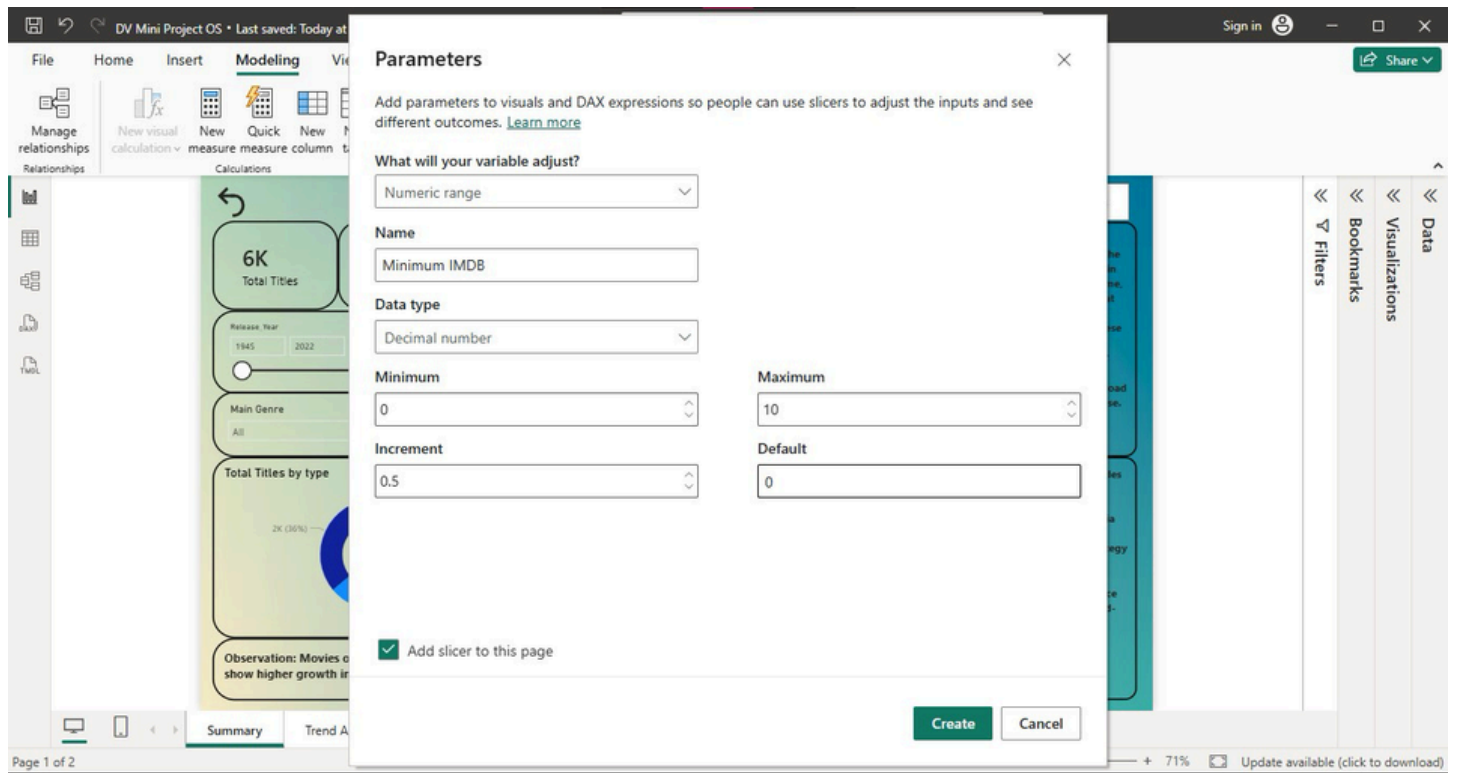
18. Add Page Navigation (buttons for smooth flow)

- Page navigation buttons are added in the Power BI dashboard to create a smooth and structured flow between different report pages, such as the **Summary Dashboard** and **Trend Analysis Dashboard**. These buttons improve user experience by allowing quick and easy movement across pages without relying on tabs. This makes the dashboard more interactive, organized, and supports a storytelling-based presentation of insights in a logical sequence.



19. Use Parameters (what-if analysis / dynamic visuals)

- Parameters are used in the Power BI dashboard to enable **what-if analysis** and create dynamic visuals for better user interaction. In this project, parameters such as **Minimum IMDB Score** allow users to adjust value thresholds and instantly analyze how the content distribution changes based on selected criteria. This helps users explore different scenarios, perform deeper analysis, and gain flexible insights by dynamically updating visuals according to the selected parameter values.



20. Create interactive and user-friendly report design

- The Power BI dashboard is designed with an interactive and user-friendly layout to make data exploration simple and effective. It includes clear KPI cards, well-organized visuals, slicers, filters, bookmarks, and navigation buttons to improve usability and accessibility. A consistent theme, structured visual hierarchy, and clean design elements are used to enhance readability, support smooth interaction, and provide a better storytelling experience for users analyzing Netflix content data.

